

In the news...

[We Are All Mutants: Measurement Of Mutation Rate In Humans By Direct Sequencing -- ScienceDaily](#)

In total, we all carry 100-200 new mutations in our DNA.

These four mutations allowed the researchers to calculate an exact mutation rate: one in 30 million nucleotides each generation

Researchers have calculated a general rate of one mutation in each 15 to 30 million DNA letters in humans. Using next-generation sequencing, researchers sequenced part of the Y chromosome from two distant male-line relatives. Despite 13 generations of separation -- with a common male ancestor 200 years ago -- they **found only four letters that differed**. Mutation is the ultimate source of human genetic variation and has implications for both evolutionary and disease genetics.

[Soapy taste of coriander linked to genetic variants | Nature](#)



Soapy taste of coriander linked to genetic variants
Dislike of coriander/cilantro traced to genes encoding odour and taste receptors

DNA Mutations

Causes and Consequences

What is a mutation?

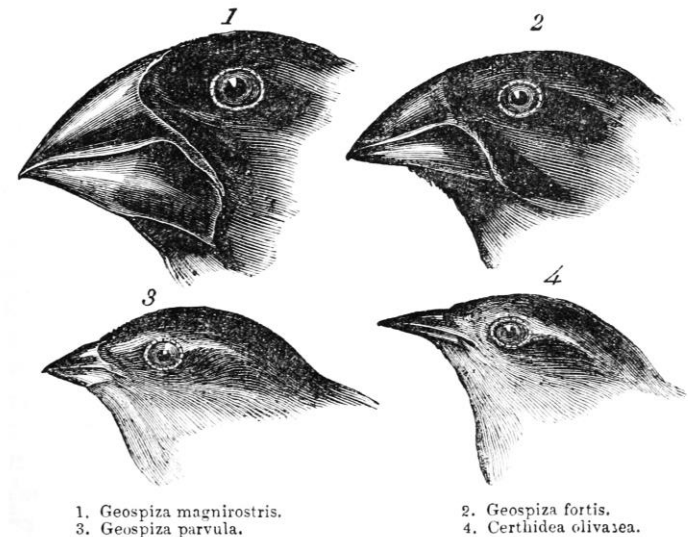
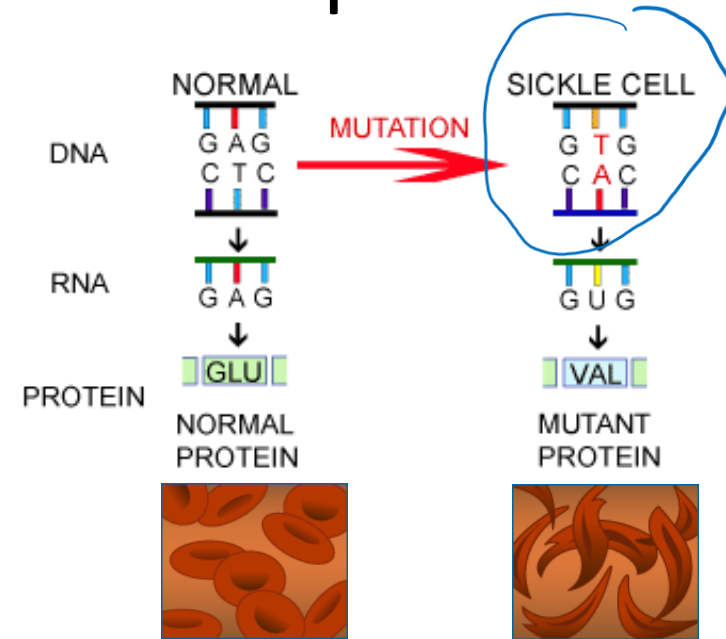
What are the consequences of mutations?

Where do mutations come from?

A mutation is a change in DNA sequence

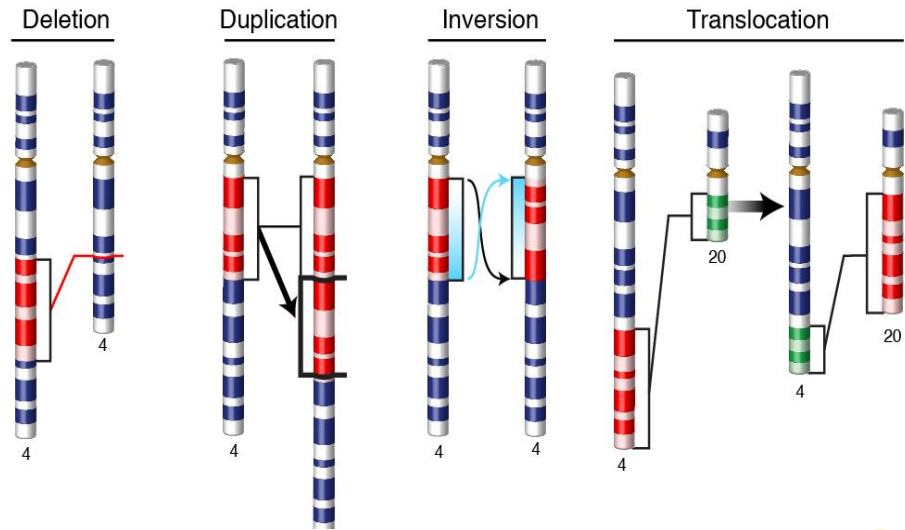
Mutations can:

- be detrimental, neutral or beneficial
- affect coding sequences or regulatory sequences
- be an important source of genetic variation
 - Germline mutations are heritable.
 - Recombination during meiosis generates different combinations of alleles
 - This allows for adaptation/evolution

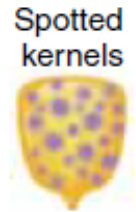
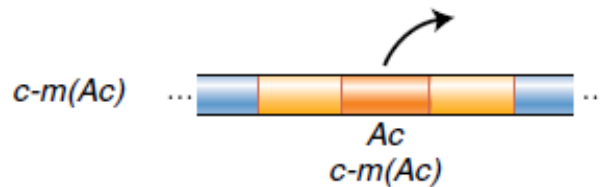


Mutations can occur on different scales

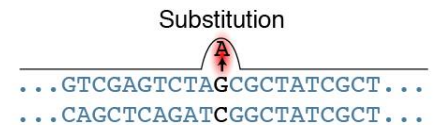
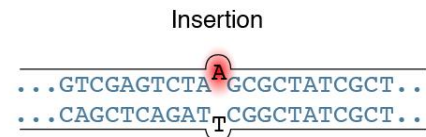
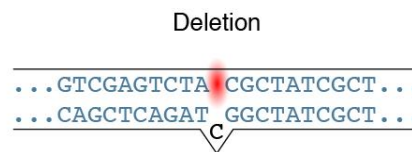
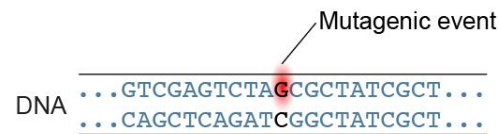
Chromosomal mutations: gain or loss of all or part of a chromosome



Insertional mutations: insertion of large regions of DNA, e.g. transposable elements



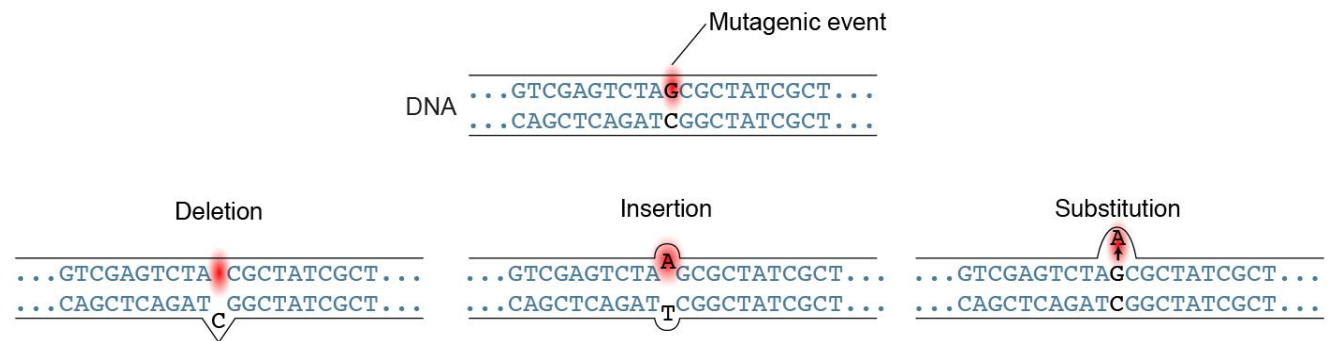
Point mutations: changes in a single nucleotide or addition/deletion of one or more nucleotides



Point mutations

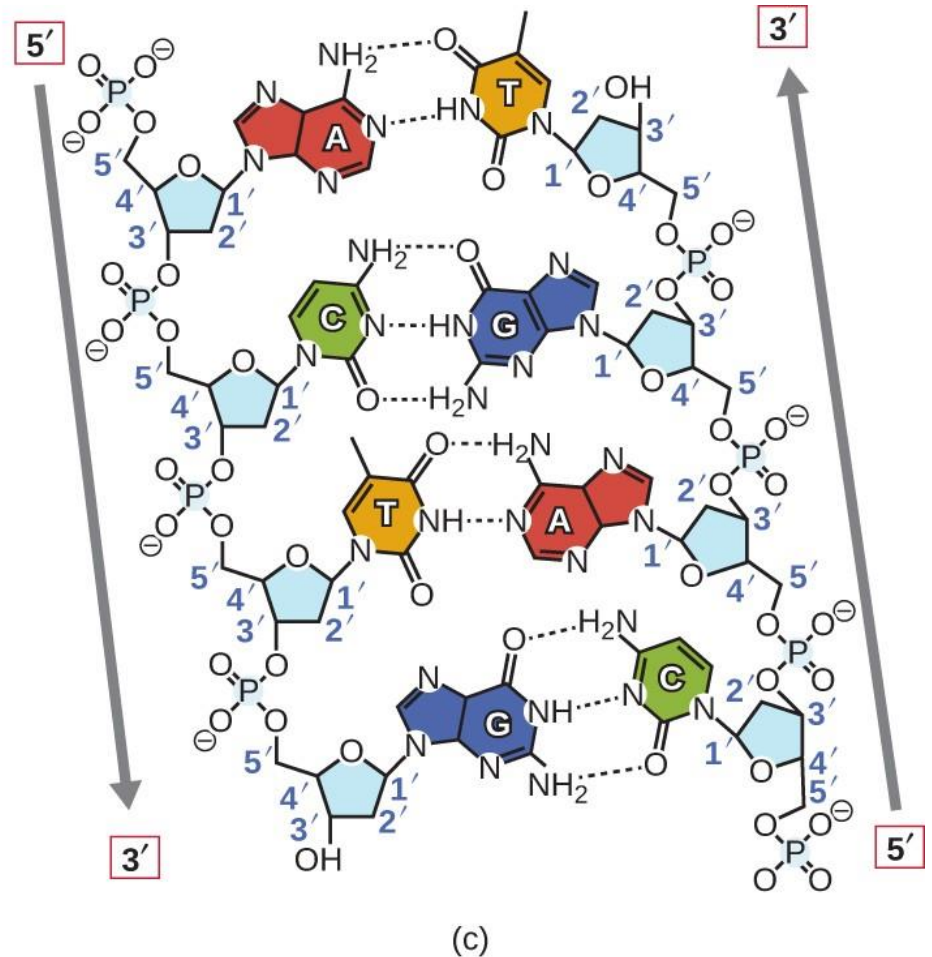
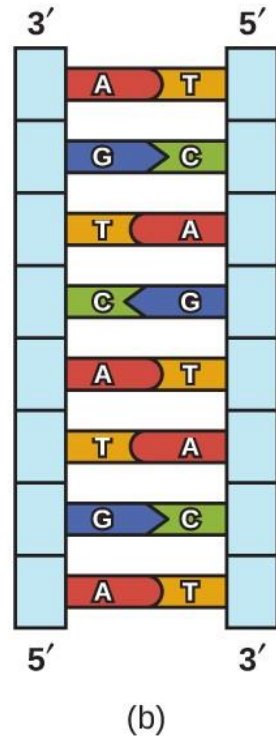
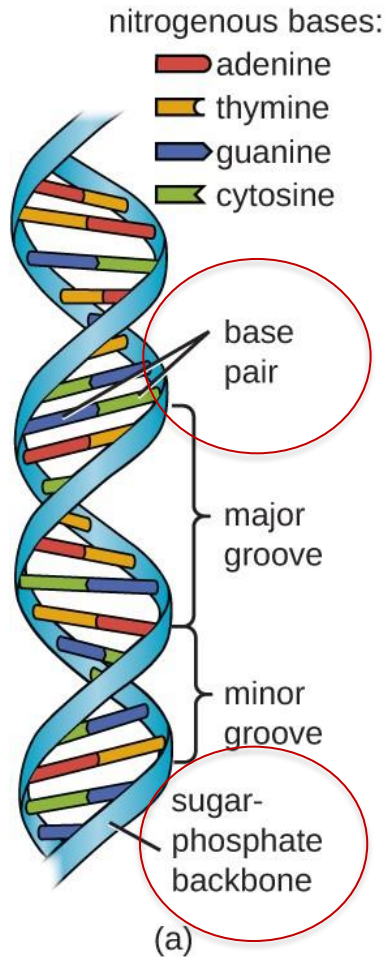
Point mutations are changes in a single nucleotide or addition/deletion of one or more nucleotides.

- How do point mutations affect gene function?
- How do they occur?

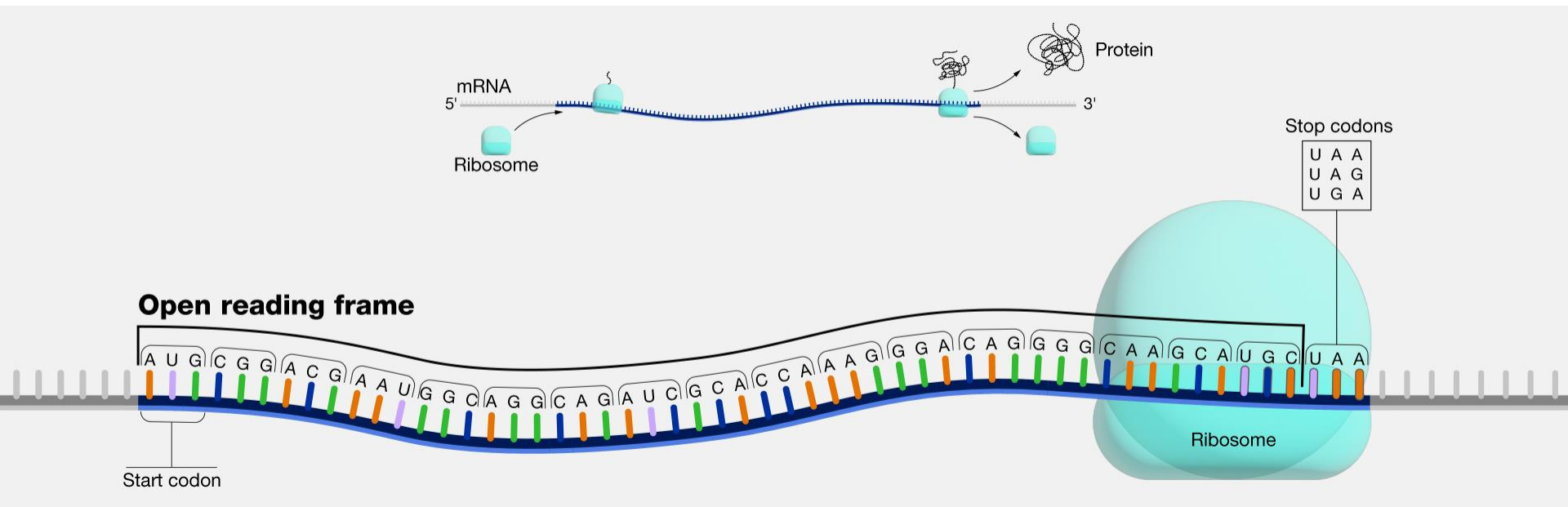


Quick DNA Refresher:

DNA structure



Quick DNA Refresher: open reading frames (ORFs)



Quick Refresher: The Genetic Code

Amino Acids			Codons						
Alanine	Ala	A	GCA	GCC	GCG	GCU			
Cysteine	Cys	C	UGC	UGU					
Aspartic acid	Asp	D	GAC	GAU	GAC	GAU			
Glutamic acid	Glu	E	GAA	GAG					
Phenylalanine	Phe	F	UUC	UUU					
Glycine	Gly	G	GGA	GCG	GGG	GGU			
Histidine	His	H	CAC	CAU					
Isoleucine	Ile	I	AUA	AUC	AUU				
Lysine	Lys	K	AAA	AAG					
Leucine	Leu	L	UUA	UUG	CUA	CUC	CUG	GUU	
Methionine	Met	M	AUG						
Asparagine	Asn	N	AAC	AAU					
Proline	Pro	P	CCA	CCC	CCG	CCU			
Glutamine	Gln	Q	CAA	CAG					
Arginine	Arg	R	AGA	AGG	CGA	CGC	CGG	CGU	
Serine	Ser	S	AGC	AGU	UCA	UCC	UCG	UCU	
Threonine	Thr	T	ACA	ACC	ACG	ACU			
Valine	Val	V	GUA	GUC	GUG	GUU			
Tryptophan	Trp	W	UGG						
Tyrosine	Tyr	Y	UAC	UAU					

Point mutations can be categorized based on their effect on the DNA sequence.

Base substitutions

Transition

purine → purine

pyrimidine → pyrimidine

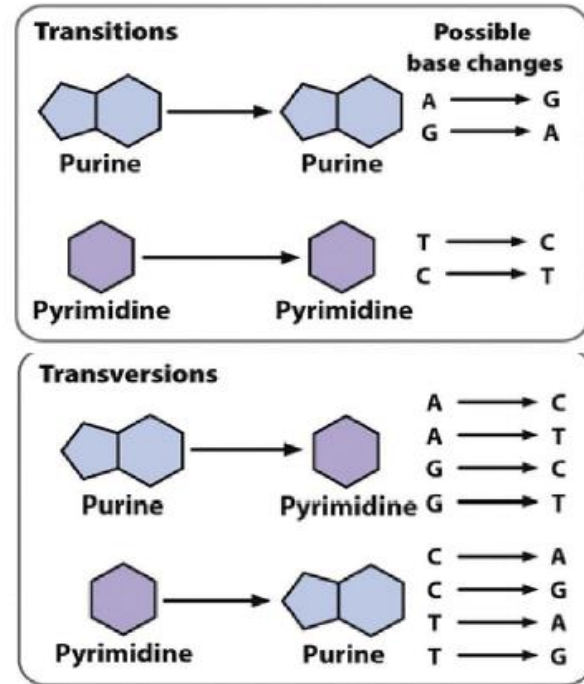
e.g. GC → AT

Transversion

pyrimidine → purine

purine → pyrimidine

e.g. CG → TA



Base additions and deletions (indels)

AAGACTCCT → AAGA**G**CTCCT
AA**G**ACTCCT → AA**A**CTCCT

Point mutations can be categorized based on their effect on mRNA coding potential.

Synonymous mutations (also called silent mutations)
change the sequence but not the amino acid it encodes

Nonsynonymous mutations (also called missense mutations)

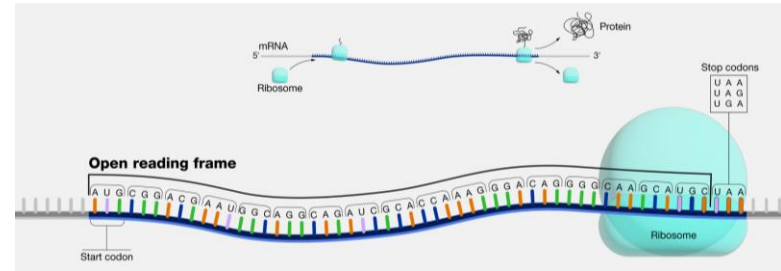
change the sequence of a codon to one that codes for a different amino acid.

Conservative mutations: mutant codon encodes a chemically similar amino acid

(e.g. lys-to-arg)

Non-conservative mutations: mutant codon encodes a chemically different amino acid

(e.g. lys-to-thr)



Nonsense mutations

change the sequence of a codon that codes for an amino acid into one that stops translation (i.e. UAA, UAG, or UGA).

Effect depends on distance from the 3' end of the ORF

Can trigger nonsense-mediated decay (NMD), which completely degrades the mRNA

Frameshift mutations

Caused by an insertion or deletion

Change the translation reading frame for all codons downstream of the mutation

Consequences of point mutations in open reading frames

		Single base substitution									
		No mutation	Silent mutation (synonymous)	Missense mutation (nonsynonymous)				Nonsense mutation			
				Conservative		Nonconservative					
DNA	CAT AAG CAG AGT ACT GTA TTC GTC TCA TGA	CAT AA A CAG AGT ACT GTA TTT GTC TCA TGA	CAT A GG CAG AGT ACT GTA T CC GTC TCA TGA	CAT A CG CAG AGT ACT GTA T GC GTC TCA TGA	CAT T AG CAG AGT ACT GTA A TC GTC TCA TGA						
mRNA	CAU AAG CAG AGU ACU	CAU AA A CAG AGU ACU	CAU A GG CAG AGU ACU	CAU A CG CAG AGU ACU	CAU U AG CAG AGU ACU						
Protein	His Lys Gln Ser Thr	His Lys Gln Ser Thr	His Arg Gln Ser Thr	His Thr Gln Ser Thr	His Stop						

Consequences of point mutations in open reading frames

		Single base insertion or base deletion (indel)	
No mutation		Insertion (frameshift)	Deletion (frameshift)
DNA	CAT TGC GAC AAG GAT AGT ACT CCT GTA ACG CTG TTC CTA TCA TGA GGA	CAT G TG CGA CAA GGA TAG TAC TCC T GTA C AC GCT GTT CCT ATC ATG AGG A ↓ ↑	CAT T GCG ACA AGG ATA GTA CTC CT GTA A CGC TGT TCC TAT CAT GAG GA ↑ ↓
mRNA	CAU UGC GAC AAG GAU AGU ACU CCU	CAU GUG CGA CAA GGA UAG UAC UCC U	CAU GCG ACA AGG AUA GUA CUC CU
Protein	His Cys Asp Lys Val Ser Thr Pro	His Val Arg Gln Gly Stop	His Ala Thr Arg Ile Val Leu

These are effects of point mutations on protein **sequence**.

What about the effect of point mutations on gene/protein **function**?

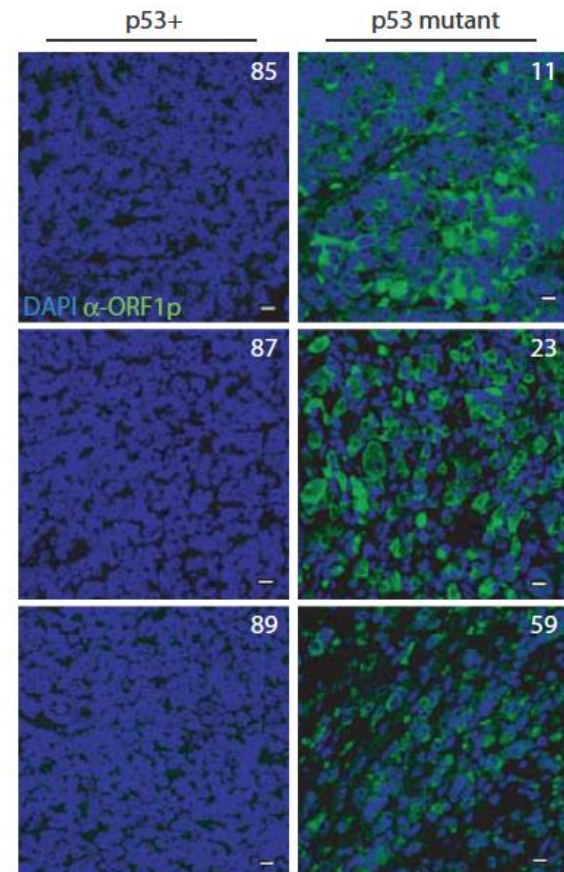
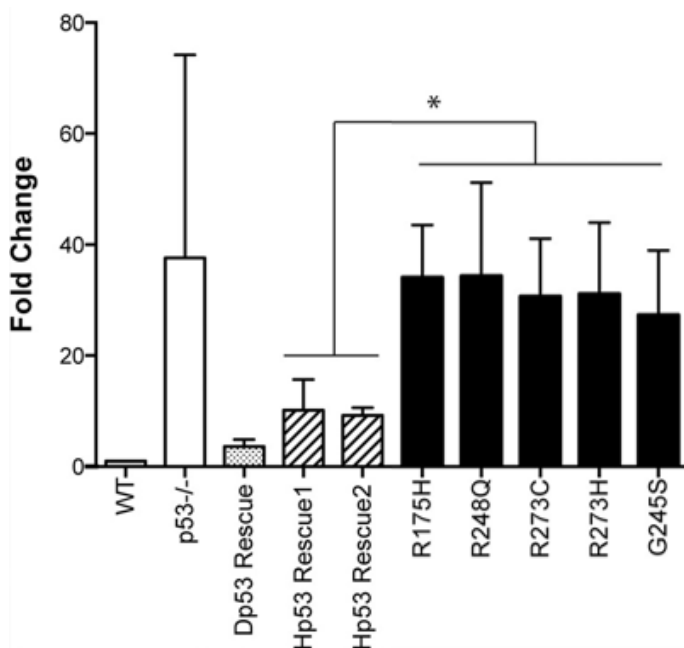
Point mutations can be categorized based on their effect on gene function.

- Wild type function (mutation has no effect on gene function)
- Loss-of-function (gene function is partially or completely lost)
 - HYPOMORPHIC: Weak/partial – protein retains some function or is produced at a reduced level
 - NULL: Protein is non-functional or not produced
- Gain-of-function (gene function is increased or novel)
 - HYPERMORPHIC: Protein is hyperactive
 - ECTOPIC: More protein made or made in wrong time/place
 - NEOMORPHIC: Protein gains new function

Mutations can be in coding or non-coding regions

Example: loss-of-function allele

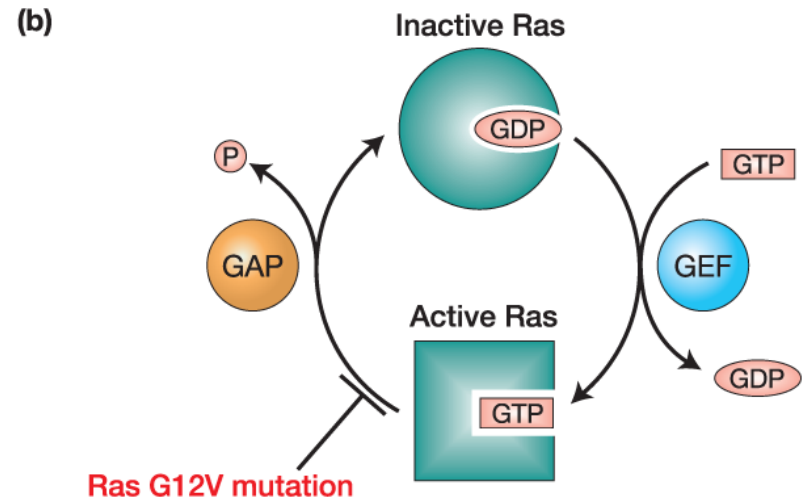
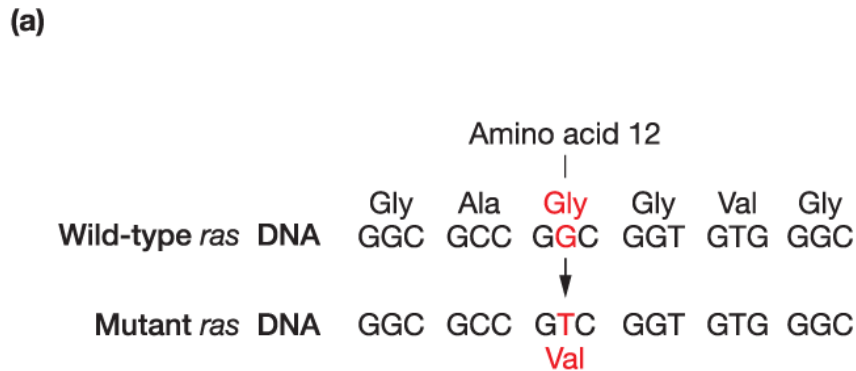
Mutant p53 alleles associated with human cancers typically contain mutations that reduce or eliminate their function.



Example: gain-of-function allele

Mutant alleles of the *ras* gene that associated with human cancers typically contain mutations that make them constitutively active.

A point mutation in the *ras* ORF affects Ras protein function



Effect on the gene

Effect on the gene product

Effect on the phenotype

- *The uncontrolled cell proliferation resulting from the mutant Ras protein can lead to cancer.*
- *Approximately 30% of human cancers are driven by mutations in Ras.*

Example: Point mutations with a neutral or positive effect

‘Happy evolutionary accident’: How crayfish living underground may have gotten their bright colors

A new study suggests the trait may have happened randomly and not as a survival tactic



In a study published in July 2024, in the *Proceedings of the Royal Society B*, an ecologist at West Liberty University, and colleagues propose that the colorful hues of crayfish evolved not for a specific purpose—such as to attract mates or warn predators—but instead purely by chance.

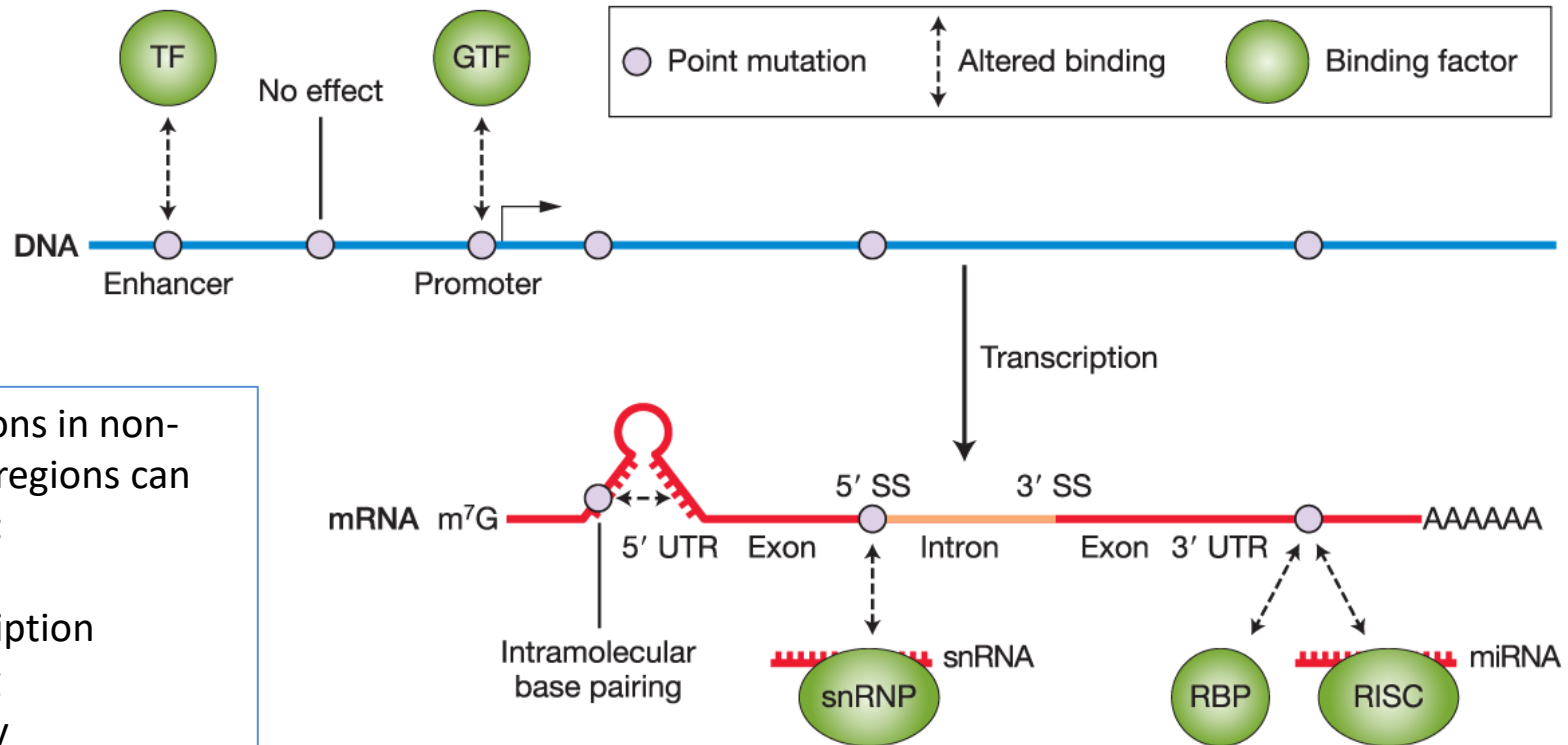
The lactose tolerance mutation arose randomly (as all mutations do), but once it arose, it had a distinct advantage in these populations.



Mutations that keep the lactase gene permanently switched on are common among modern Europeans — but not among their ancestors. Results suggest that as late as 5000 BC most ancient Europeans could not have digested milk as adults — and that they only later evolved into milk-drinking societies.

Sequences in non-coding regions can be important for gene function

Consequences of point mutations in noncoding regions



Mutations in non-coding regions can disrupt:

Transcription
Splicing
Stability
Translation
Function
etc.

Sequences in non-coding regions can be important for gene function

Point mutations can alter mRNA splicing*

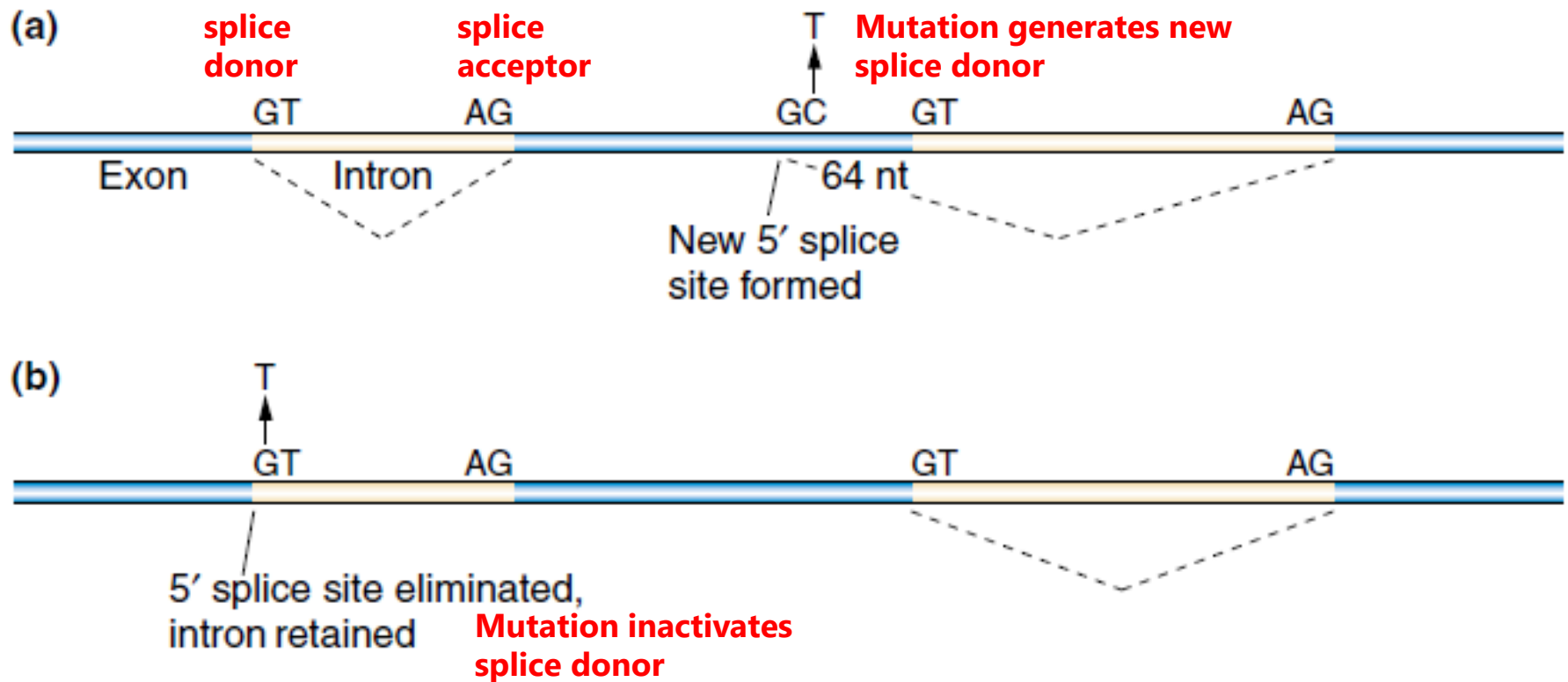
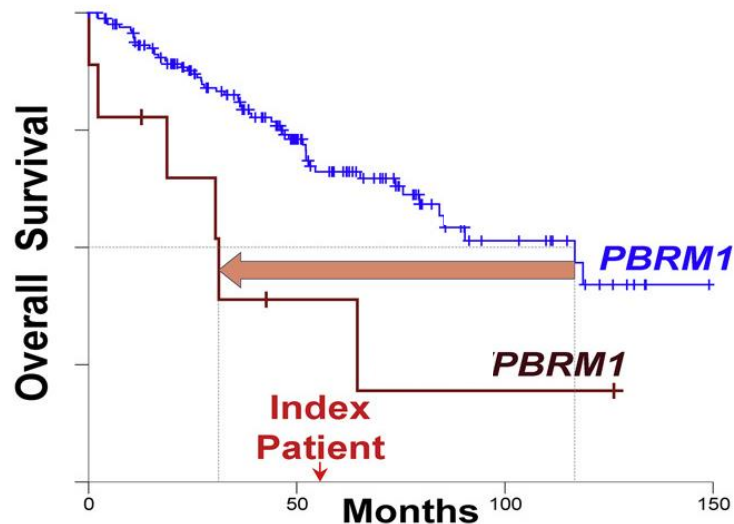


Figure 16-3
Introduction to Genetic Analysis, Eleventh Edition
© 2015 W. H. Freeman and Company

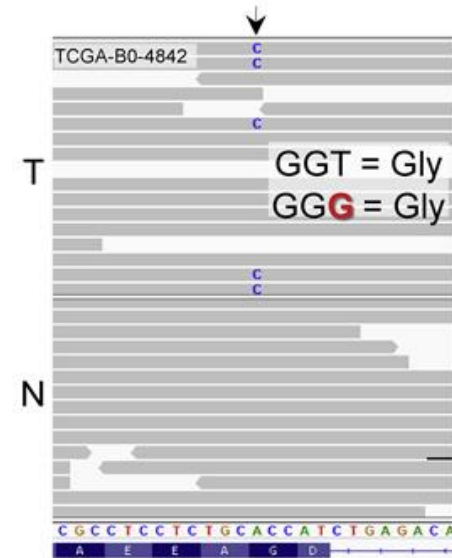
* can be in non-coding or coding regions

Example: A synonymous mutation can cause a loss of gene function, though rarely.

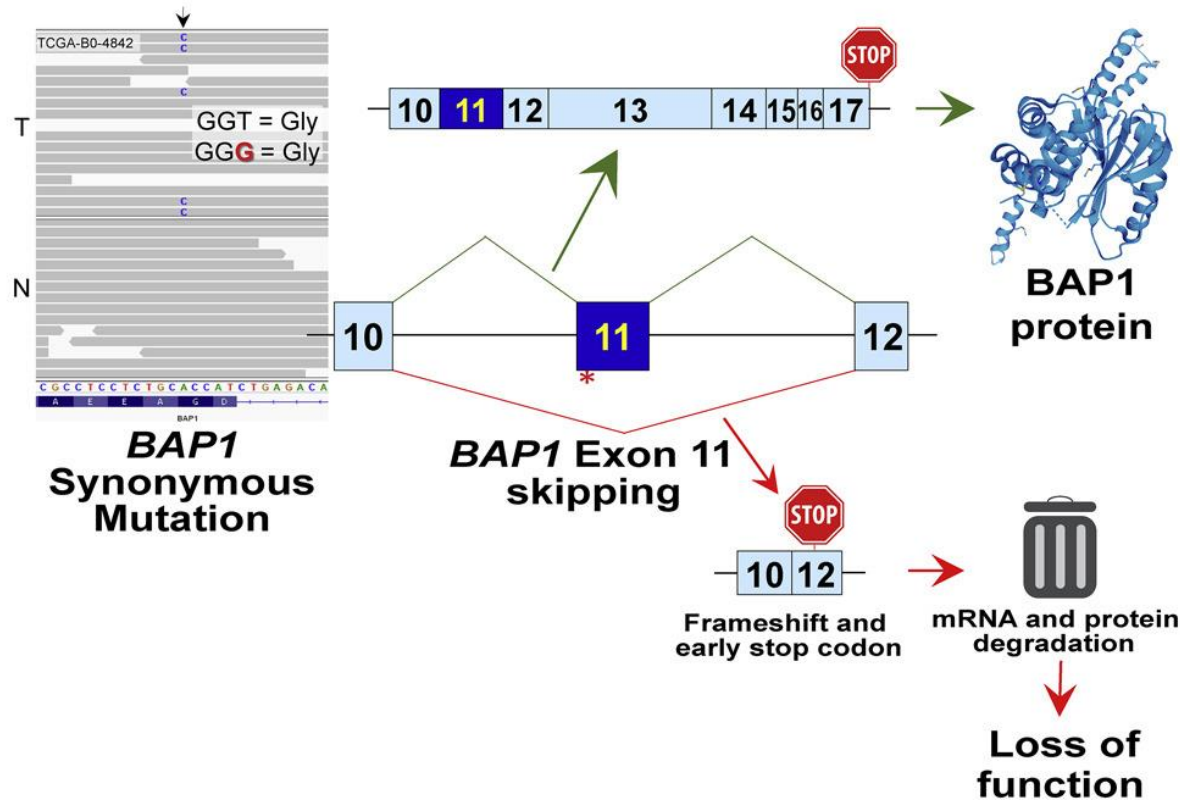
Observation: A kidney cancer patient with a mutation in the PBRM1 gene had a much worse outcome than expected. Why?



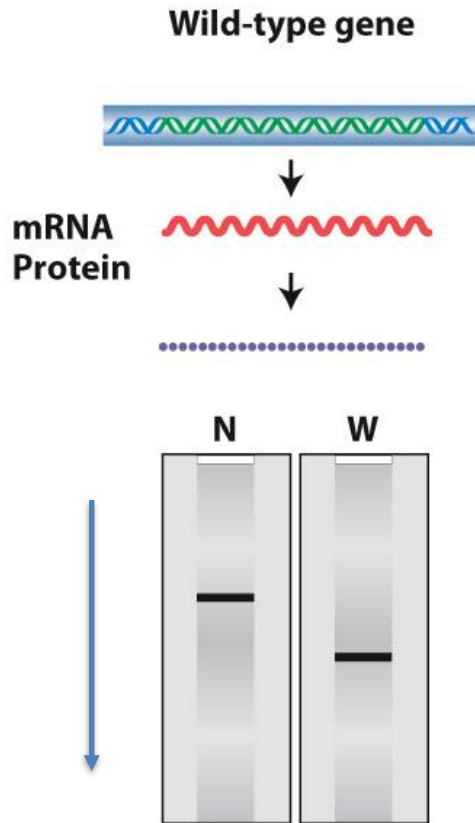
Hypothesis: This patient has a mutation in another gene, in addition to PBRM1, that makes the cancer outcome worse.



Example: A synonymous mutation in a BAP1 exon leads to loss of BAP1 function through exon skipping.



Detecting effects of point mutations on a gene's mRNA and protein products



Sometimes, but not always, mutations lead to detectable changes in the mRNA or protein generated from the gene.

Reminders:

N: Northern blots are for looking at RNA.

W: Western blots are for looking at protein.

Electrophoretic mobility (position on gel) reflects size, band intensity reflects amount present.

Consequences of point mutations within genes

NOTE: an early stop codon can lead to mRNA degradation (via “nonsense mediated decay”) which can lead to no RNA and protein being produced at all.

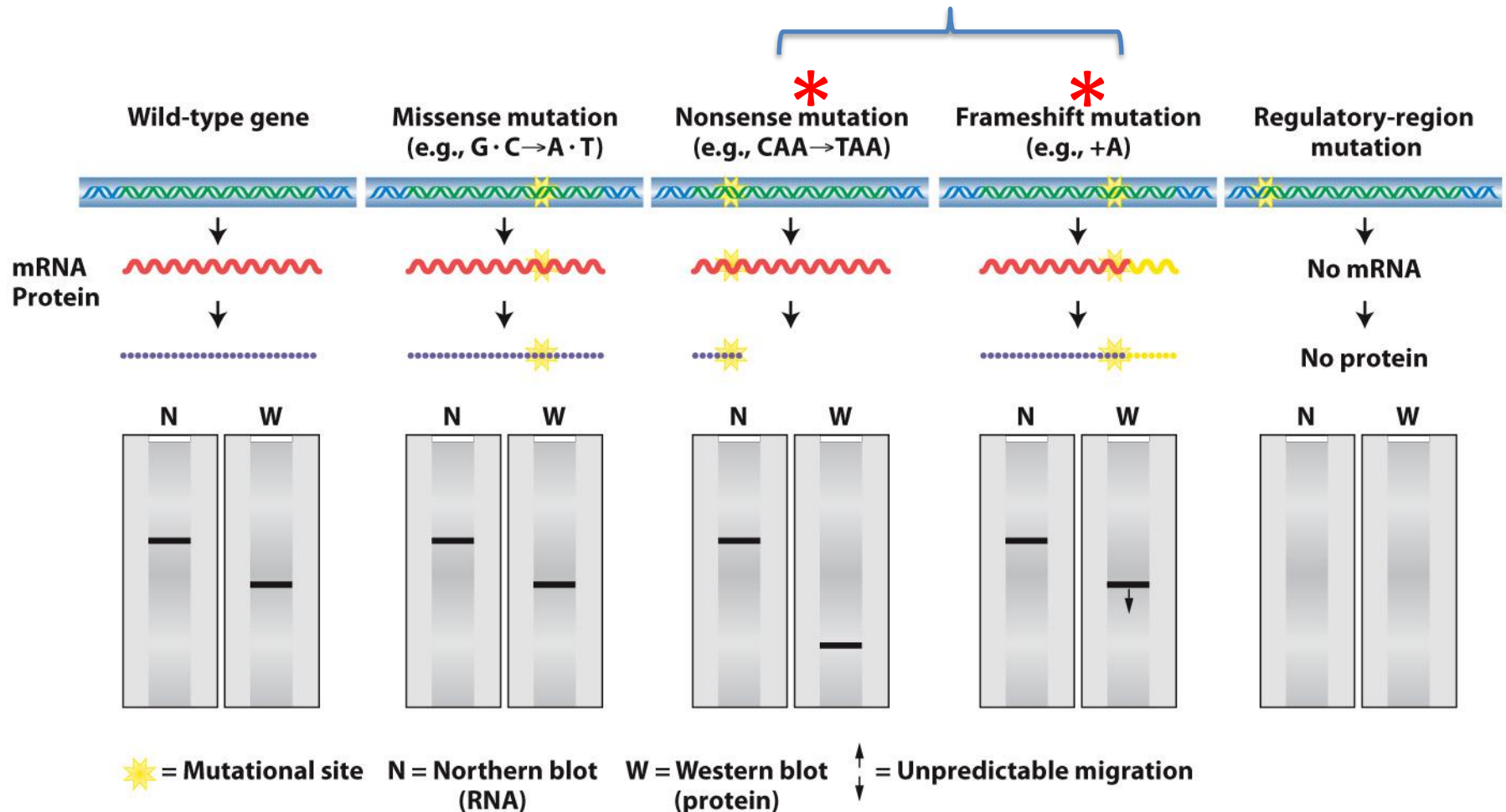
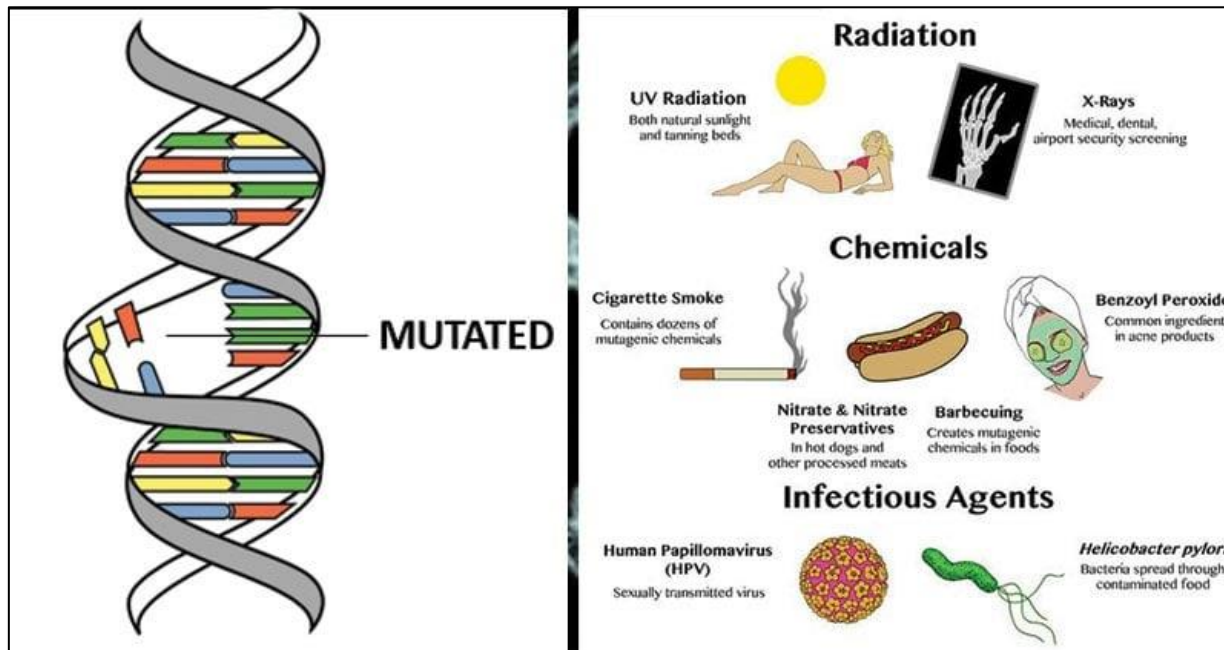


Figure 16-4
Introduction to Genetic Analysis, Eleventh Edition
 © 2015 W. H. Freeman and Company

Mutations can be spontaneous or induced

spontaneous: occur naturally, arise in all cells

induced: caused by exposure to ionizing radiation, mutagenic chemicals, etc., which increase the rate at which mutations occur



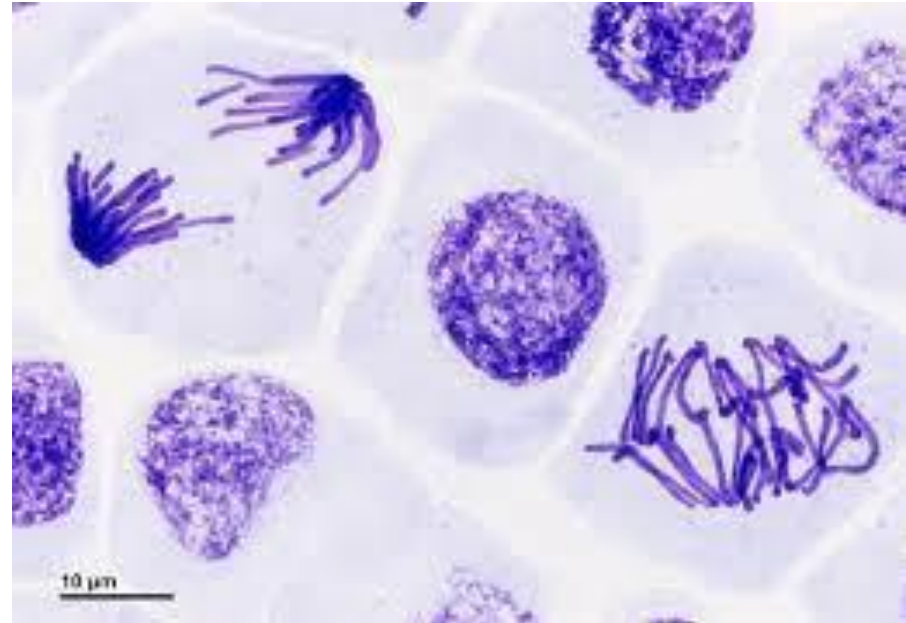
Origins of spontaneous mutations

1. Errors in DNA replication

- Base mispairing
- Strand slippage

2. Chemical changes to the DNA (which lead to errors in DNA replication)

- Depurination
- Deamination



DNA polymerase error rate = 10^{-8}
Times 3×10^9 base pairs in the genome =
~300 mutations per mitosis

Luckily, >99% are fixed by DNA repair mechanisms!

Researchers have calculated a mutation rate of one in 30 million nucleotides each generation.

Therefore, in total, we each carry 100-200 new mutations in our DNA.

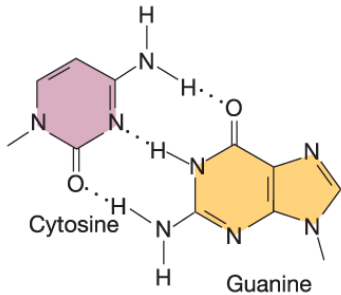
DNA replication errors

Spontaneous mutations can be caused by errors in base pairing

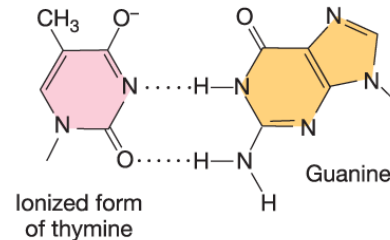
Tautomerization and ionization of bases leads to mismatched base pairing

WHEN?

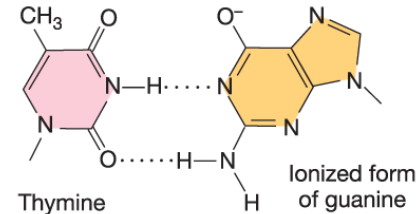
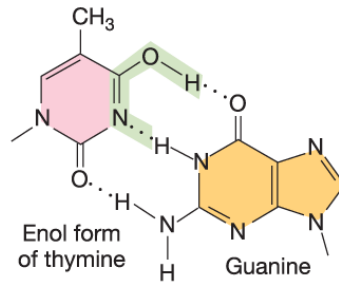
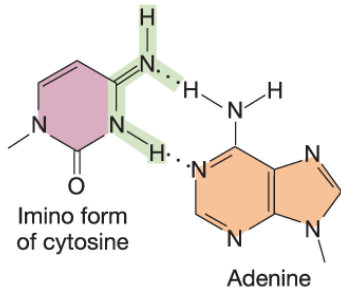
(a) Normal base pairs



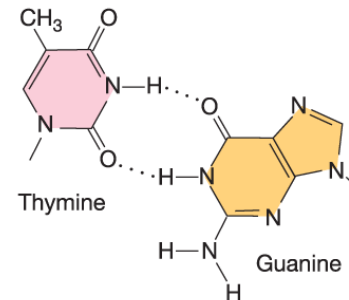
(c) Ionized base pairs



(b) Tautomeric base pairs



(d) Wobble base pair

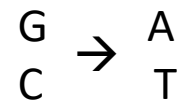
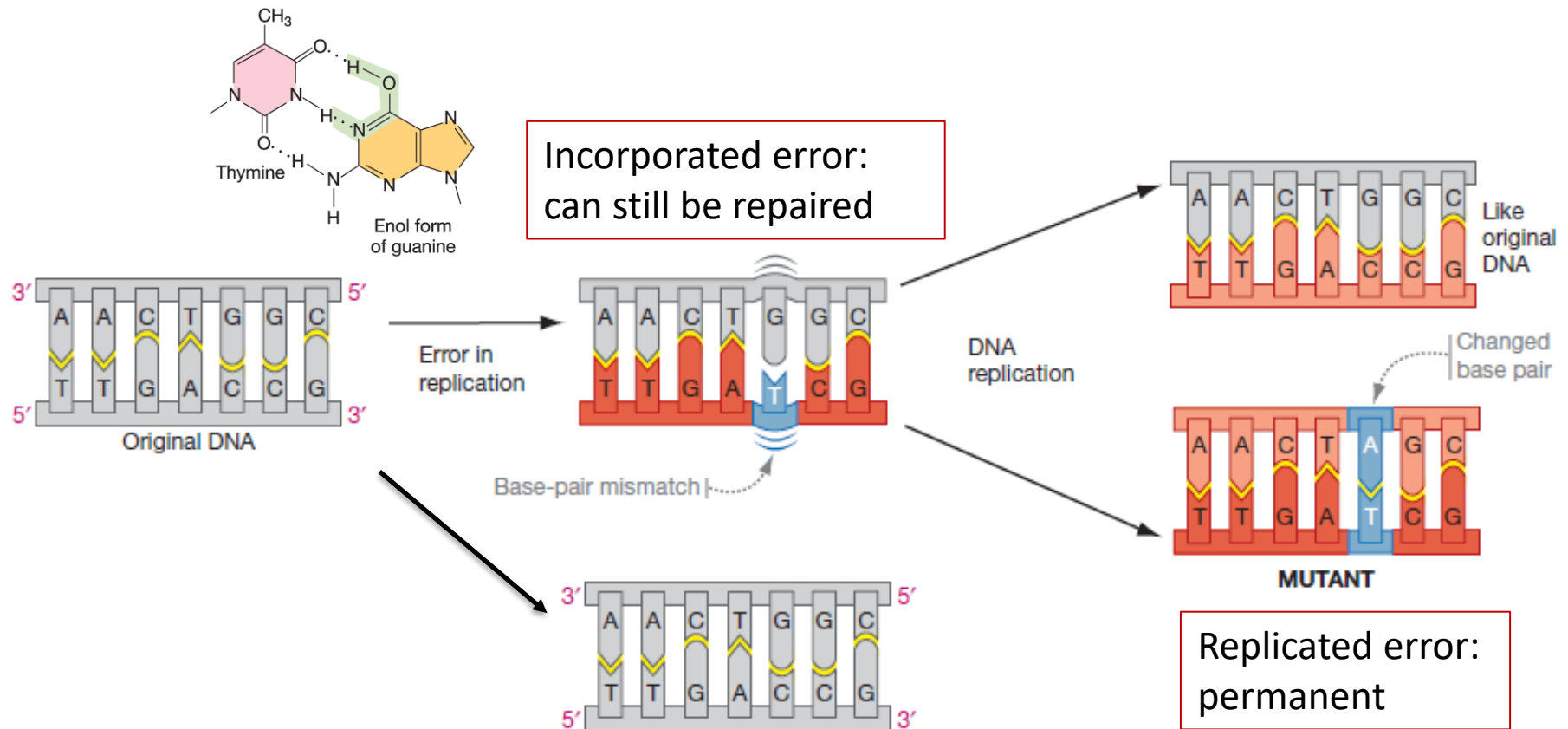


Tautomers: isomers that differ in the positions of their atoms and in the bonds between the atoms. The forms are in equilibrium.

If proofreading does not occur, the mismatches lead to transition mutations

Spontaneous mutations:

Errors in DNA replication due to errors in base pairing



Spontaneous mutations: strand slippage during replication

Insertion

Direction of DNA replication →

5' — **CGTTT**
3' — **GCAAAAACGTAC** —

Newly synthesized strand slips
Extra base loops out

5' — **CG TTT**
3' — **GC AAAAACGTAC** —

Loop stabilized by
repetitive sequences

5' — **CG TTTTGCATG**
3' — **GC AAAAACGTAC** —

Next round of
replication

5' — **CGTTT TTTTGCATG** —
3' — **GCAAAAACGTAC** —

5' — **CGTTT TTTTGCATG** —
3' — **GCAAAAACGTAC** —

T • A base pair added

Deletion

Direction of DNA replication →

5' — **CTGAGAGA**
3' — **GACTCTCTCTCTGCA** —

Template strand slips
Extra bases loop out

5' — **CT GAGAGA**
3' — **GA CTCTCTCTGCA** —

Loop stabilized by
repetitive sequences

5' — **CT GAGAGAGACGT**
3' — **GA CTCTCTCTGCA** —

Next round of
replication

5' — **CTGAGAGAGACGT** —
3' — **GACTCTCTCTGCA** —

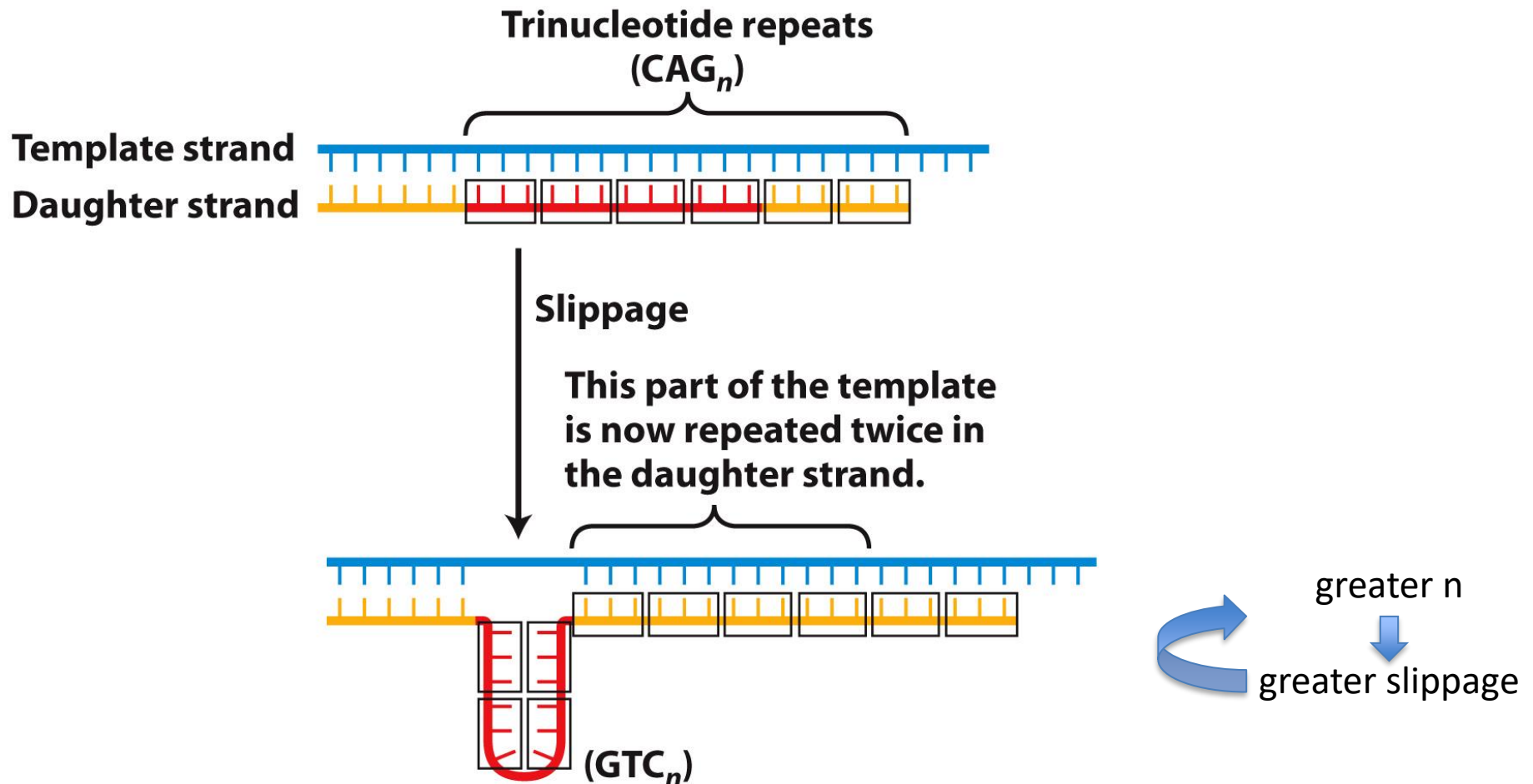
5' — **CTGAGAGAGAGACGT** —
3' — **GACTCTCTCTCTGCA** —

G • C and A • T base pairs deleted

Strand slippage during DNA replication leads to indels.
Whether slippage generates an insertion or deletion depends on which strand loops out.

Spontaneous mutations: strand slippage during replication

Trinucleotide repeat disorders arise from expansion of repeated sequences.



Strand slippage is the cause of many “trinucleotide repeat diseases”

Here: List of diseases caused by expanded CAG tracts

Since CAG encodes glutamine (Q), these are known as “polyQ” diseases.

Expanded polyQ tracts lead to abnormal folding, which leads to protein aggregation and neural degeneration.

Disease	Gene	Locus	Protein	CAG repeat size	
				Normal	Disease
Spinobulbar muscular atrophy (Kennedy disease)	<i>AR</i>	Xq13–21	Androgen receptor (AR)	9–36	38–62
Huntington’s disease	<i>HD</i>	4p16.3	Huntingtin	6–35	36–121
Dentatorubral-pallidoluysian atrophy (Haw–River syndrome)	<i>DRPLA</i>	12p13.31	Atrophin-1	6–35	49–88
Spinocerebellar ataxia type 1	<i>SCA1</i>	6p23	Ataxin-1	6–44 ^a	39–82
Spinocerebellar ataxia type 2	<i>SCA2</i>	12q24.1	Ataxin-2	15–31	36–63
Spinocerebellar ataxia type 3 (Machado–Joseph disease)	<i>SCA3 (MJD1)</i>	14q32.1	Ataxin-3	12–40	55–84
Spinocerebellar ataxia type 6	<i>SCA6</i>	19p13	α _{1A} -voltage-dependent calcium channel subunit	4–18	21–33
Spinocerebellar ataxia type 7	<i>SCA7</i>	13p12–13	Ataxin-7	4–35	37–306

Example: trinucleotide repeat expansion in non-coding region of the FMR1 gene causes Fragile X Syndrome

Methylation of trinucleotide repeats blocks transcription of *FMR1*

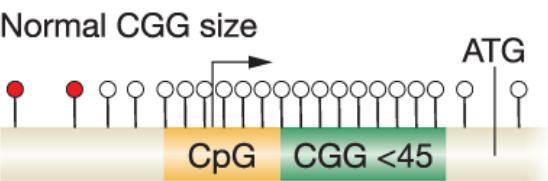
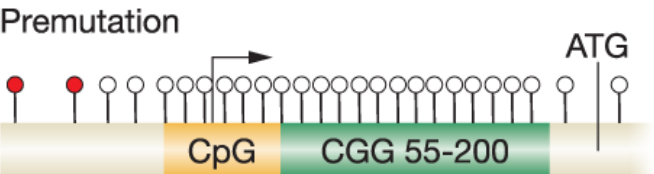
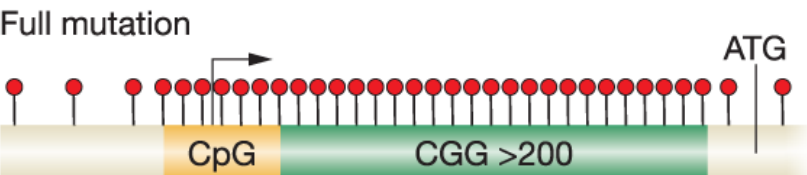
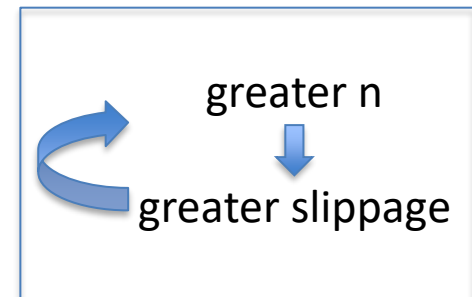
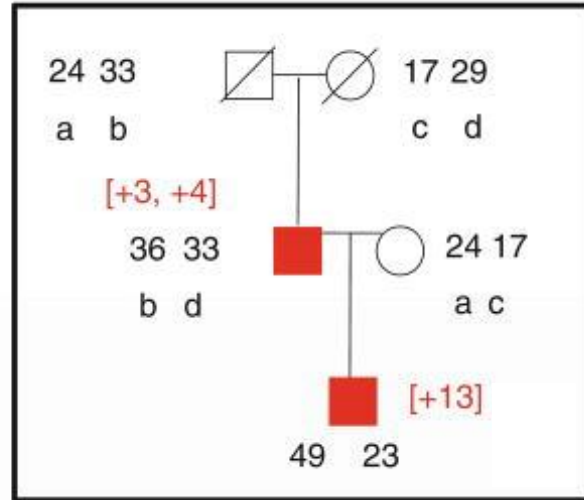
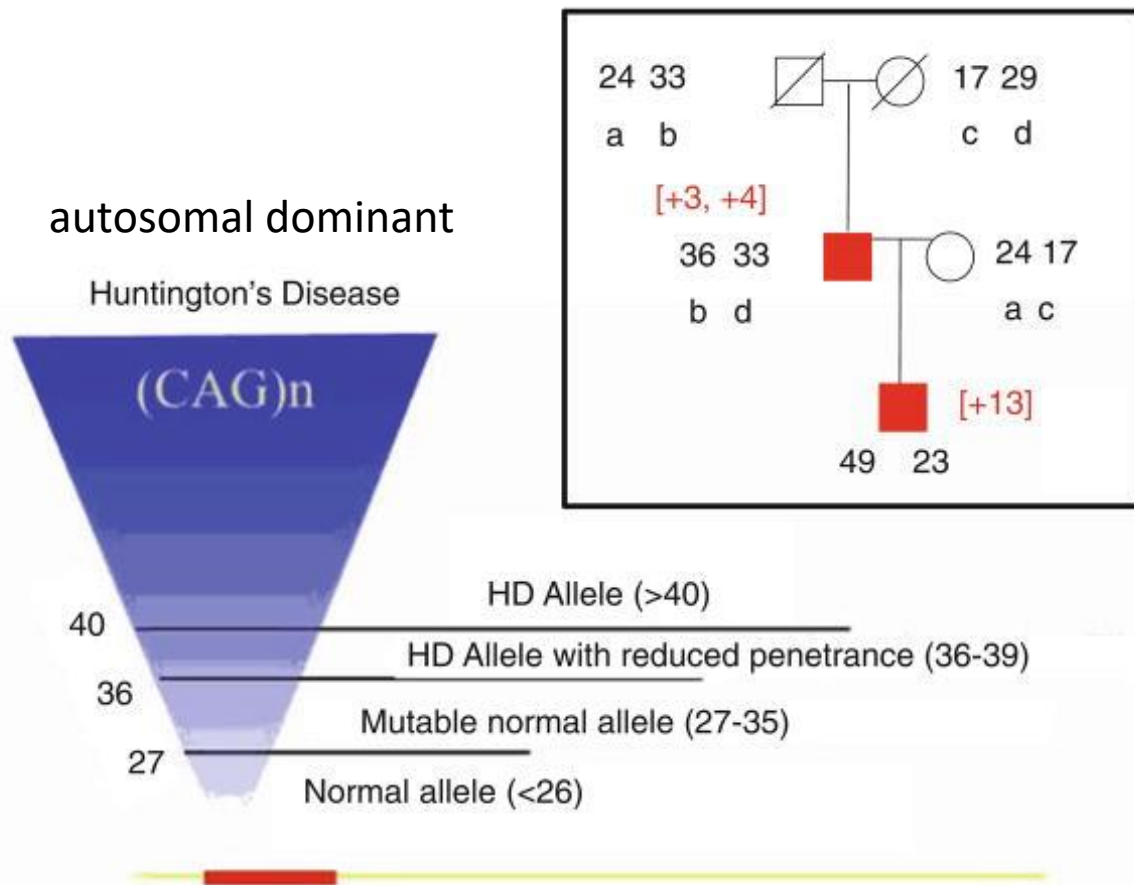
<i>FMR1</i> gene	Phenotype	Transmission	Methylation	Transcription
Normal CGG size 	Normal	Stable	No	Yes
Premutation 	Largely normal	Unstable, prone to expansion	No	Yes
Full mutation 	Affected	Unstable, prone to expansion	Yes	No

Fig 15.8. The number of trinucleotide CGG repeats in the 5' UTR of *FMR1* affects phenotype, the probability of repeat expansion, CpG methylation, and transcription. Red and white lollipops indicate methylated and unmethylated cytosines, respectively, CpG indicates a CpG island, the arrow indicates the transcription start site, and ATG indicates the translation start site.

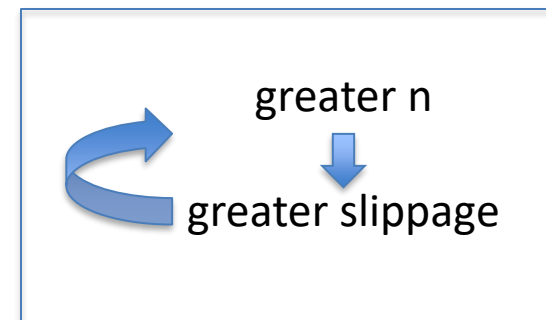
BIOL 202 W2025 | © LA Nilson



Example: trinucleotide repeat expansion in coding region of the HTT gene causes Huntington's Disease



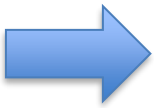
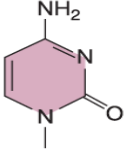
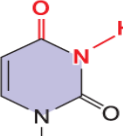
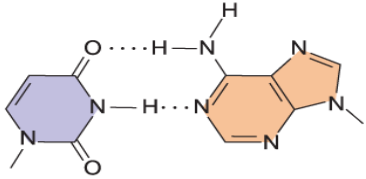
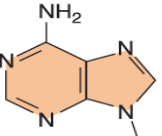
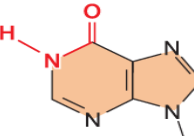
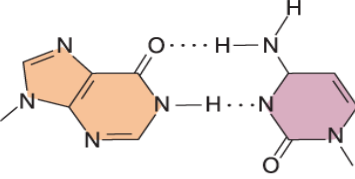
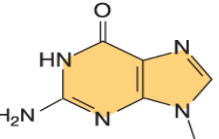
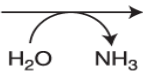
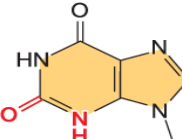
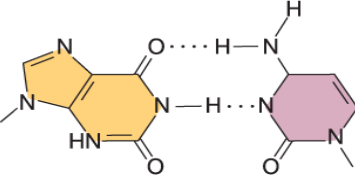
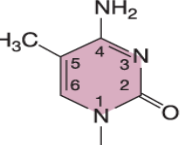
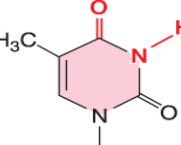
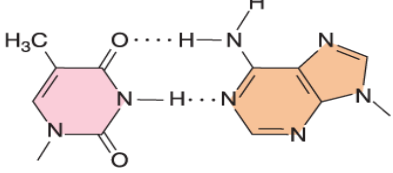
← Number of repeats
in each HTT allele
(HTT: the gene affected in
Huntington's Disease)



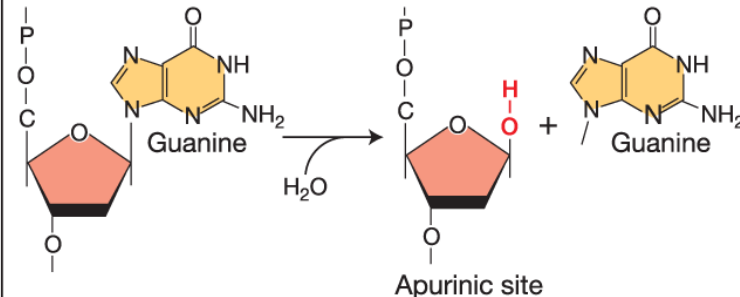
Spontaneous chemical changes to DNA

Spontaneous mutations can be caused by spontaneous chemical changes to DNA

Spontaneous mutations caused by the cellular environment

DNA damage type	Original base	Modified base	Base-pairing interaction	Consequences
Deamination 	 Cytosine	  Uracil	 Uracil Adenine	$C \cdot G \rightarrow T \cdot A$ Transition
	 Adenine	  Hypoxanthine	 Hypoxanthine Cytosine	$A \cdot T \rightarrow G \cdot C$ Transition
	 Guanine	  Xanthine	 Xanthine Cytosine	$G \cdot C \rightarrow G \cdot C$ No change
	 5-Methylcytosine	  Thymine	 Thymine Adenine	$C \cdot G \rightarrow T \cdot A$ Transition

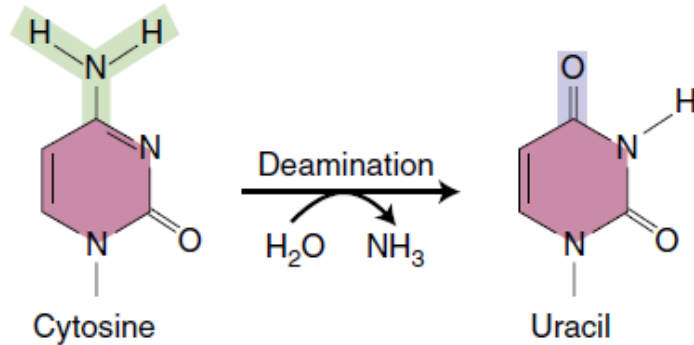
Example of a spontaneous chemical change to DNA: Depurination

Spontaneous mutations caused by the cellular environment				
DNA damage type	Original base	Modified base	Base-pairing interaction	Consequences
Depurination			None	Potentially mutagenic Blocked DNA replication and transcription

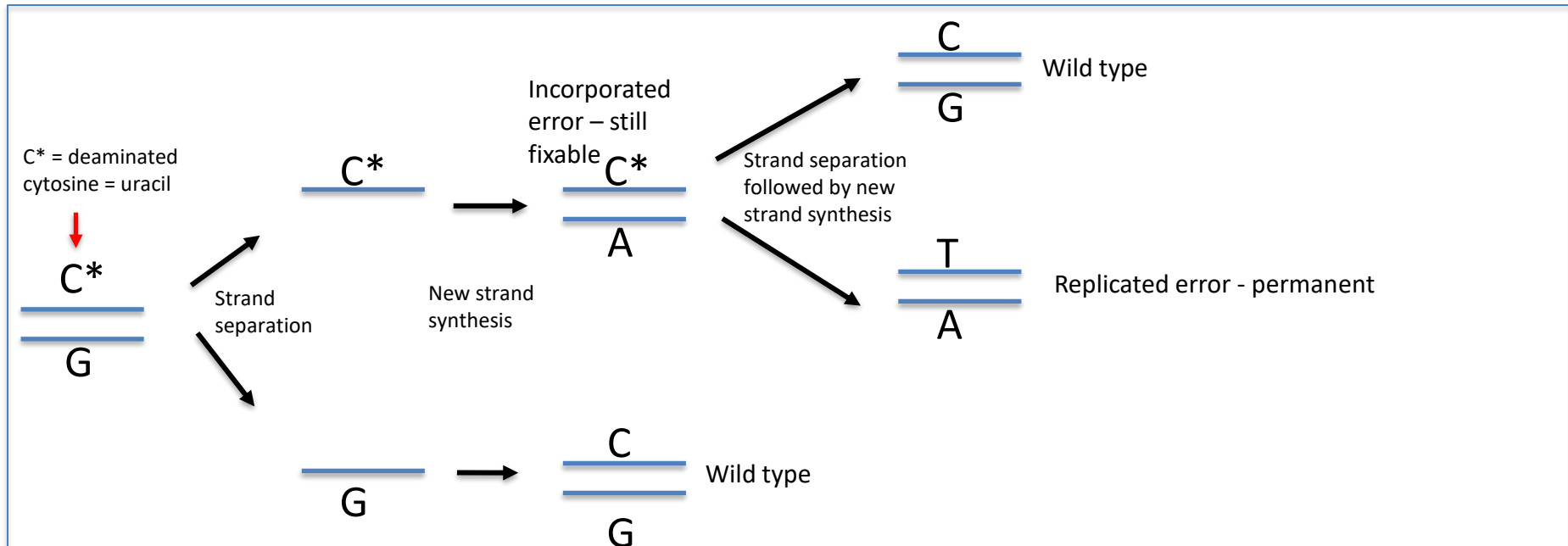
Can lead to sequence errors during replication.
10,000 purines lost per cell in a day! (usually repaired)

Example of a spontaneous chemical change to DNA: Deamination

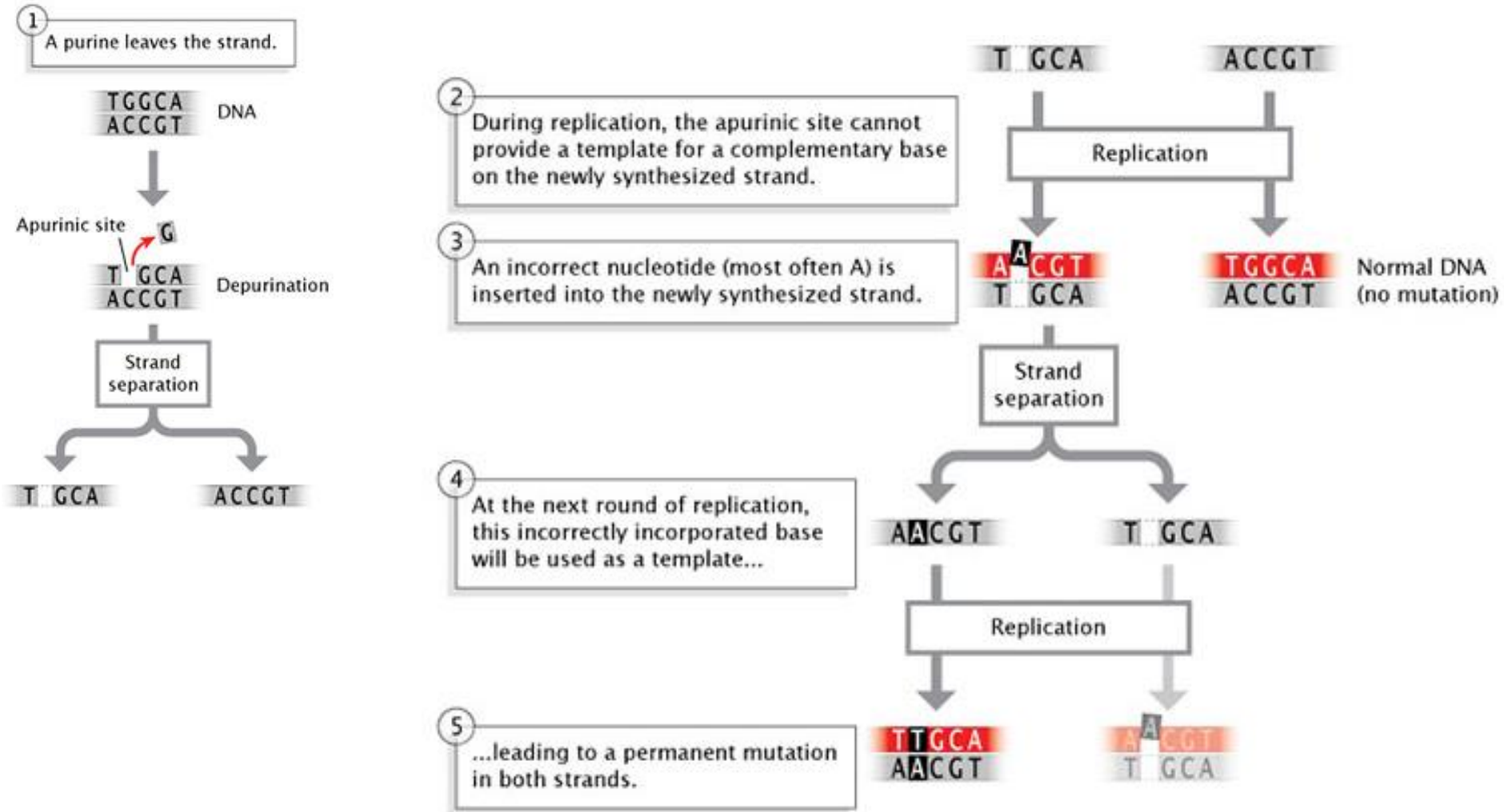
Spontaneous mutations caused by the cellular environment



conversion of a C · G pair into an T · A pair
(C · G → T · A transition)



Mispairing of mutated bases can lead to permanent DNA mutations



Molecular basis of **induced** mutations

Molecular basis of **induced** mutations

Mutagens can induce mispairing or damage/non-recognition

Radiation

UV Radiation
both natural sunlight
and tanning beds



X-Rays
medical, dental,
airport security screening

Chemicals

Cigarette Smoke
contains dozens of
mutagenic chemicals



Benzoyl Peroxide
common ingredient
in acne products

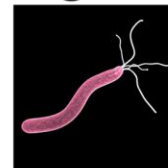


**Nitrate and Nitrite
Preservatives**
in hot dogs and
other processed meats

Barbecuing
creates mutagenic
chemicals in foods

Infectious Agents

**Human Papillomavirus
(HPV)**
sexually transmitted virus



Helicobacter pylori
bacteria spread through
contaminated food

Molecular basis of induced mutations

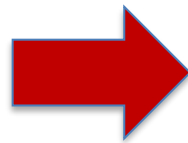
Mutagens can induce mispairing or damage/non-recognition

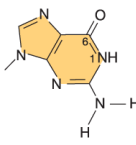
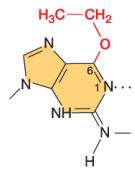
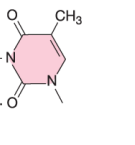
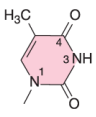
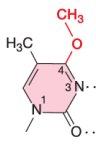
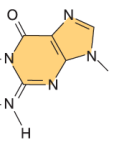
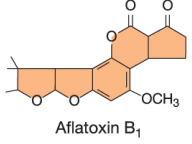
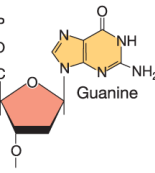
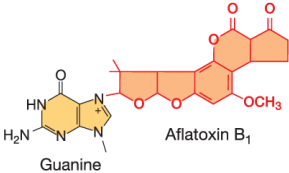
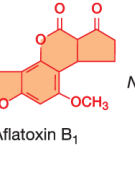
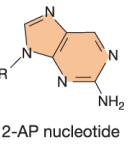
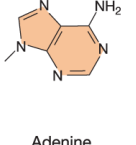
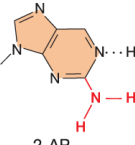
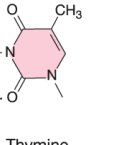
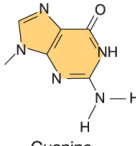
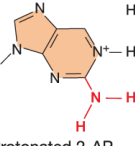
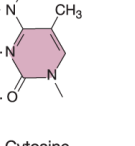
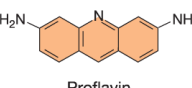
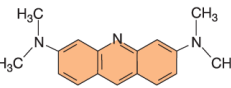
Mutagens induce mutations by at least three different mechanisms.

- *replace* a base in DNA, i.e. get incorporated instead of an actual base (base analogs)
- *alter* a base so that it mispairs with another base (e.g. alkylating agents)
- *damage* a base so that it can no longer base pair with any base (e.g. UV light, ionizing radiation)

Molecular basis of induced mutations

Some examples of
mutagens and their
effects



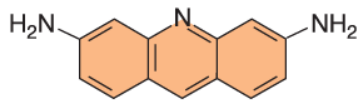
Induced mutations caused by exogenous chemical agents				
Mutagen	Original base	Modified base	Base-pairing partner	Consequences
(a) Alkylating agents <chem>CC(=O)OS(=O)(=O)C</chem> EMS <chem>CN(C)[N+](=O)[O-]</chem> MNNG	 Guanine	 O-6-ethylguanine	 Thymine	G • C → A • T Transition
	 Thymine	 O-4-methylthymine	 Guanine	T • A → C • G Transition
(b) Bulky adducts  Aflatoxin B ₁	 Guanine	 Guanine	 Aflatoxin B ₁	Potentially mutagenic Blocked DNA replication and transcription
(c) Base analogs  2-AP nucleotide	 Adenine	 2-AP	 Thymine	A • T → G • C Transition
	 Guanine	 Protonated 2-AP	 Cytosine	G • C → A • T Transition
(d) Intercalating agents  Proflavin  Acridine orange	 Nitrogenous bases Intercalated molecule			Base insertions and deletions

Molecular basis of induced mutations

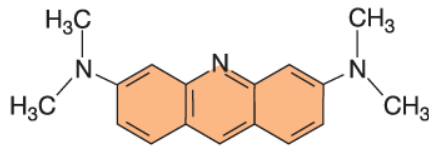
Intercalating agents

Induced mutations caused by exogenous chemical agents

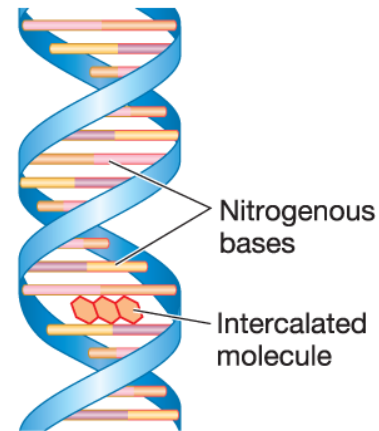
(d) Intercalating agents



Proflavin

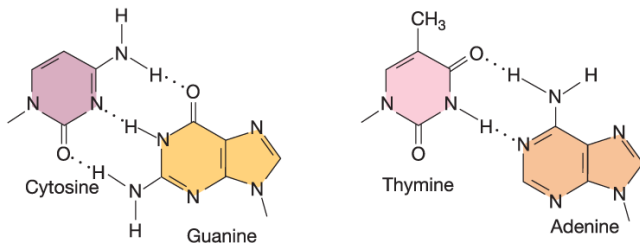


Acridine orange



Base insertions
and deletions

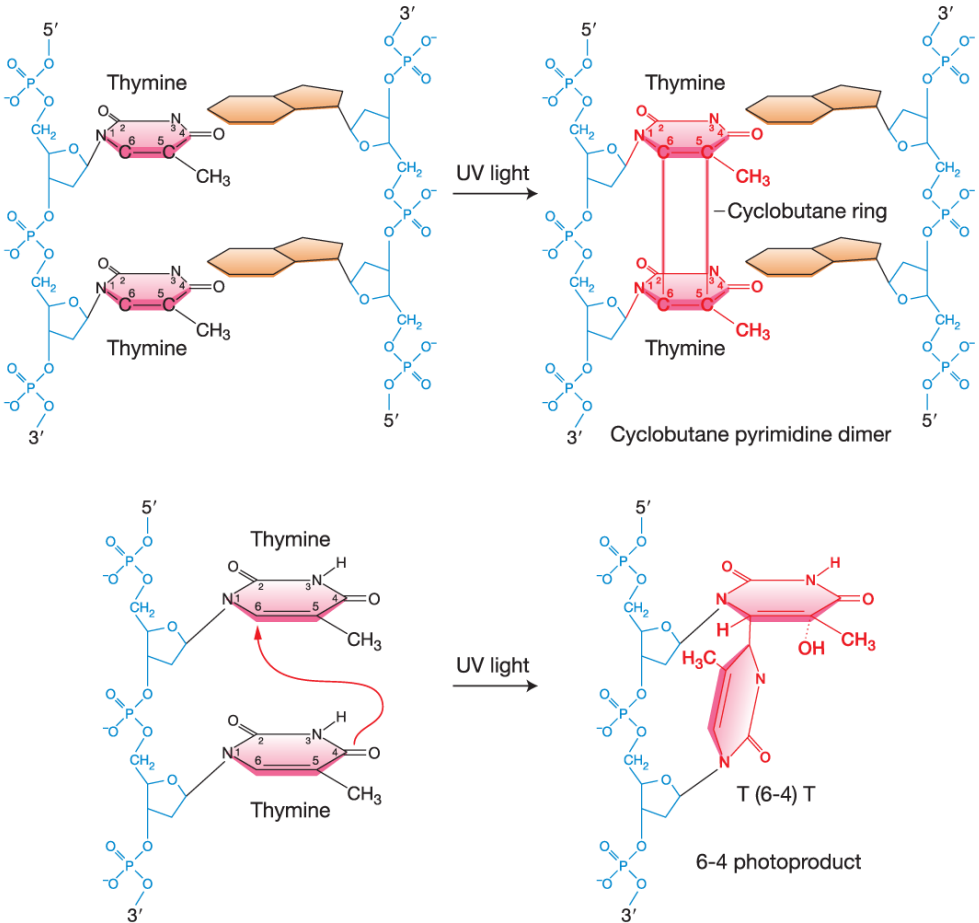
(a) Normal base pairs



- Mimic base pairs and insert into DNA double helix
- Disrupt spacing and can cause insertions or deletions of single nucleotide pairs

Molecular basis of induced mutations

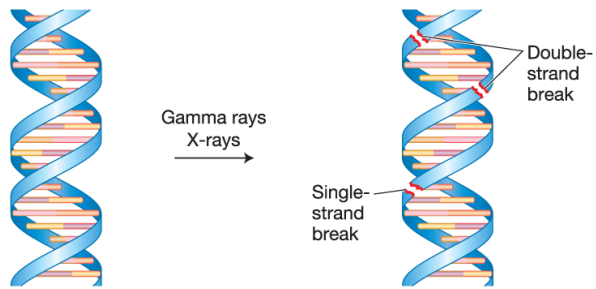
Base damage – UV light

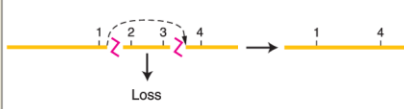
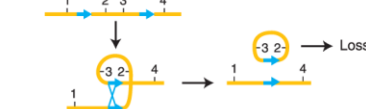
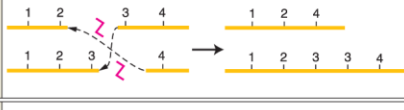
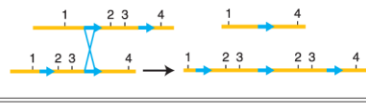
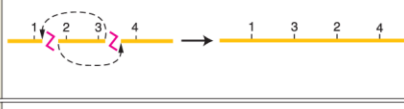
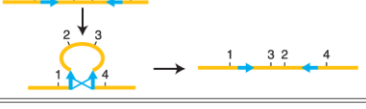
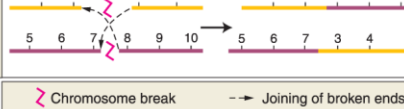
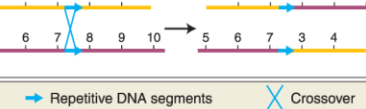
Induced mutations caused by exogenous physical agents		
Mutagen	Reaction	Consequences
(a) UV light	 Thymine Thymine UV light Cyclobutane pyrimidine dimer Thymine Thymine UV light T (6-4) T 6-4 photoproduct	Pyrimidine transitions Blocked DNA replication and transcription

- UVB light can induce covalent interactions between adjacent pyrimidines
- If not repaired, this can block DNA replication

Molecular basis of induced mutations

Base damage – ionizing radiation

Induced mutations caused by exogenous physical agents		
Mutagen	Reaction	Consequences
(b) Ionizing radiation		Various mutations due to strand break repair and oxidative damage

Origins of chromosomal rearrangements		
	Breakage and rejoining	Crossing over between repetitive DNA
Deletion		
Deletion and duplication		
Inversion		
Translocation		
 Chromosome break  Joining of broken ends  Repetitive DNA segments  Crossover		

Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

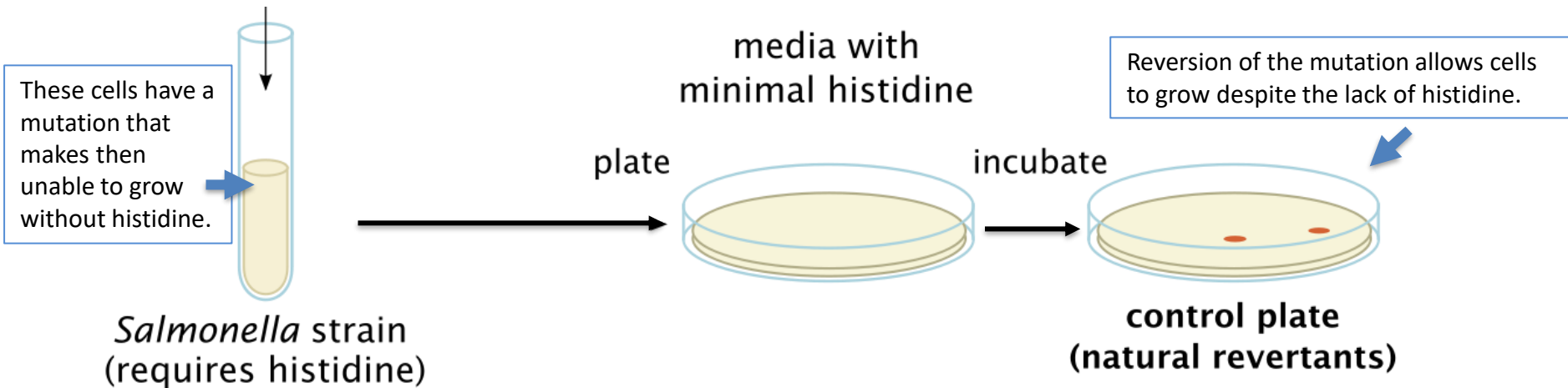
How do we know if something is a mutagen?

How do we know if something is a mutagen?

Ask if it can cause mutations! But how?

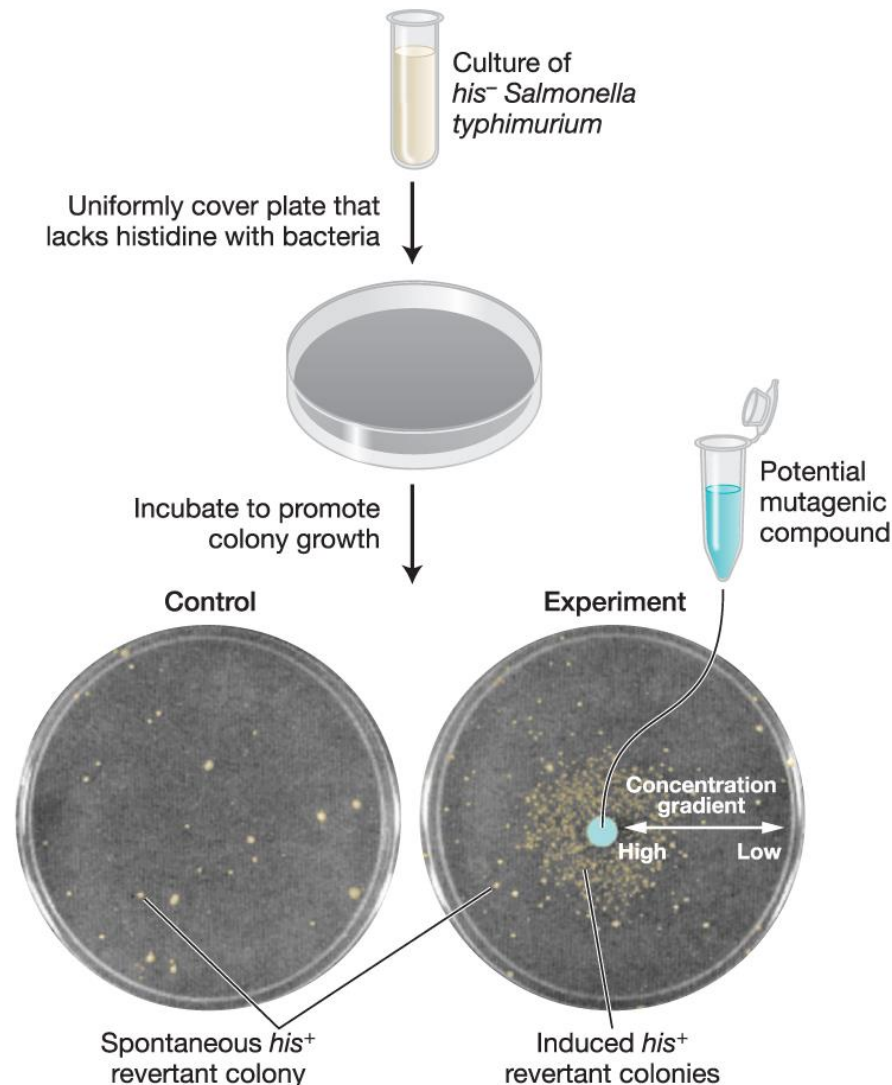
The Ames test

Starting point: a *his*-mutant strain of Salmonella bacteria, which cannot grow in the absence of histidine. Growth in the presence of only low levels of histidine means that the bacterium has undergone a spontaneous mutation that has allowed it to survive. These rare surviving colonies are called “revertants.”



Rationale behind the Ames test: if treatment of these *his*-mutant Salmonella cells with a particular compound increases the number of revertants, then that compound is a mutagen.

The Ames test for mutagenic compounds



A problem with the Ames test



Culture of
*his⁻ Salmonella
typhimurium*



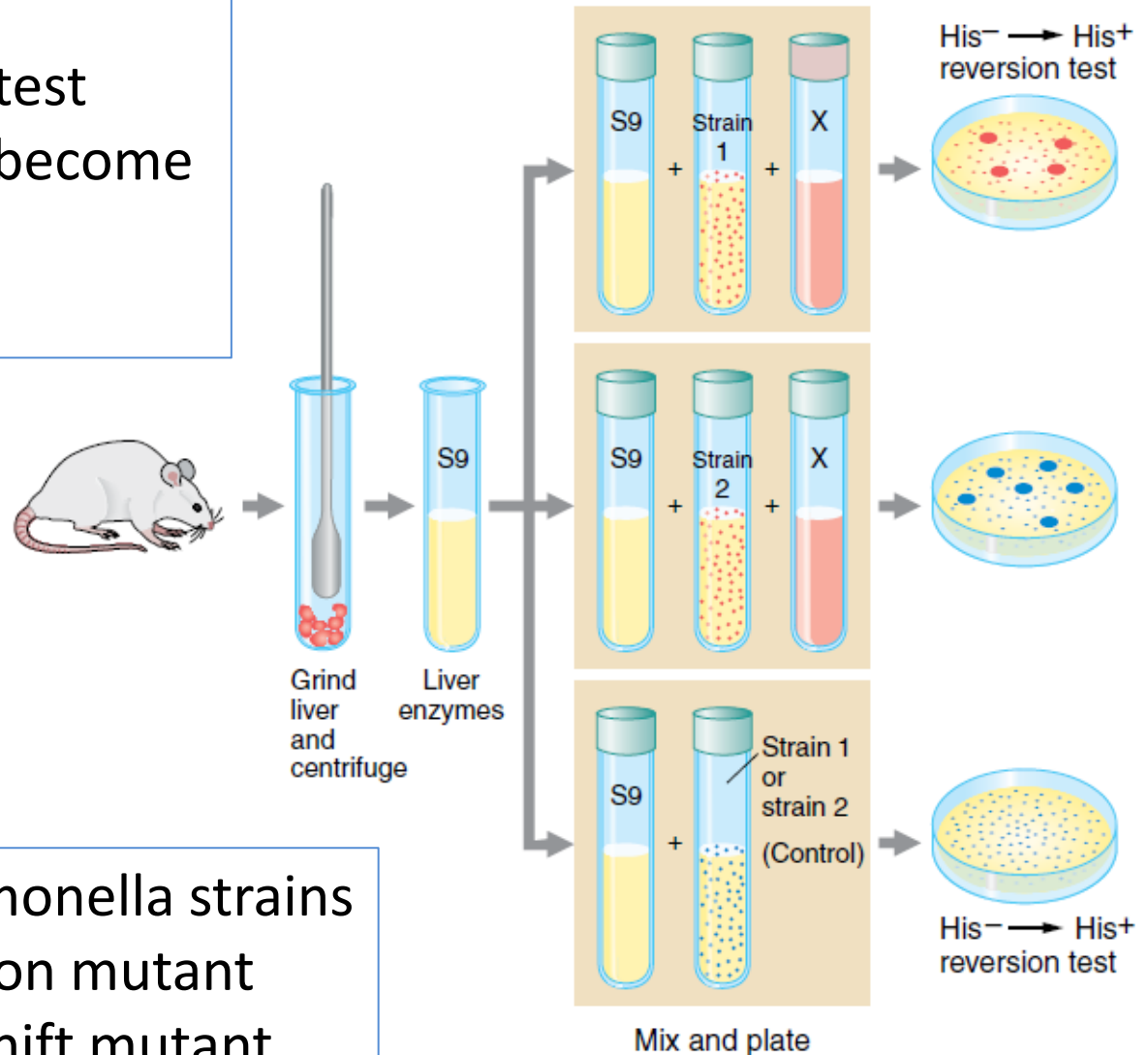
The Ames test for mutagenic compounds

Revised Ames test:

Add liver enzymes to test chemicals that might become mutagenic only when metabolized

Why?

use two different Salmonella strains
Strain 1 = His⁻ transition mutant
Strain 2 = His⁻ frameshift mutant



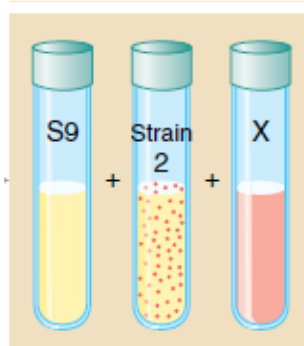
Why did the revised Ames test include these two different Salmonella strains?

		Single base substitution																								
		No mutation					Silent mutation (synonymous)				Missense mutation (nonsynonymous)				Nonsense mutation											
											Conservative		Nonconservative													
DNA		CAT	AAG	CAG	AGT	ACT	CAT	AA A	CAG	AGT	ACT	CAT	AG G	CAG	AGT	ACT	CAT	AC G	CAG	AGT	ACT	CAT	TA G	CAG	AGT	ACT
		GTA	TTC	GTC	TCA	TGA	GTA	TT T	GTC	TCA	TGA	GTA	TC C	GTC	TCA	TGA	GTA	TC G	GTC	TCA	TGA	GTA	AT C	GTC	TCA	TGA
mRNA		CAU	AAG	CAG	AGU	ACU	CAU	AA A	CAG	AGU	ACU	CAU	AG G	CAG	AGU	ACU	CAU	AC G	CAG	AGU	ACU	CAU	UA G	CAG	AGU	ACU
Protein		His	Lys	Gln	Ser	Thr	His	Lys	Gln	Ser	Thr	His	Arg	Gln	Ser	Thr	His	Thr	Gln	Ser	Thr	His	Stop			

						Single base insertion or base deletion (indel)																									
						No mutation												Insertion (frameshift)						Deletion (frameshift)							
DNA		CAT	TGC	GAC	AAG	GAT	AGT	ACT	CCT	CAT	G TG	CGA	CAA	GGA	TAG	TAC	TCC	T	CAT	T GCG	ACA	AGG	ATA	GTA	CTC	CT					
											↓									↑											
		GTA	ACG	CTG	TTC	CTA	TCA	TGA	GGA	GTA	C AC	GCT	GTT	CCT	ATC	ATG	AGG	A	GTA	A CGC	TGT	TCC	TAT	CAT	GAG	GA					
mRNA		CAU	UGC	GAC	AAG	GAU	AGU	ACU	CCU	CAU	GUG	CGA	CAA	GGA	UAG	UAC	UCC	U	CAU	GCG	ACA	AGG	AUA	GUA	CUC	CU					
Protein		His	Cys	Asp	Lys	Val	Ser	Thr	Pro	His	Val	Arg	Gln	Gly	Stop				His	Ala	Thr	Arg	Ile	Val	Leu						

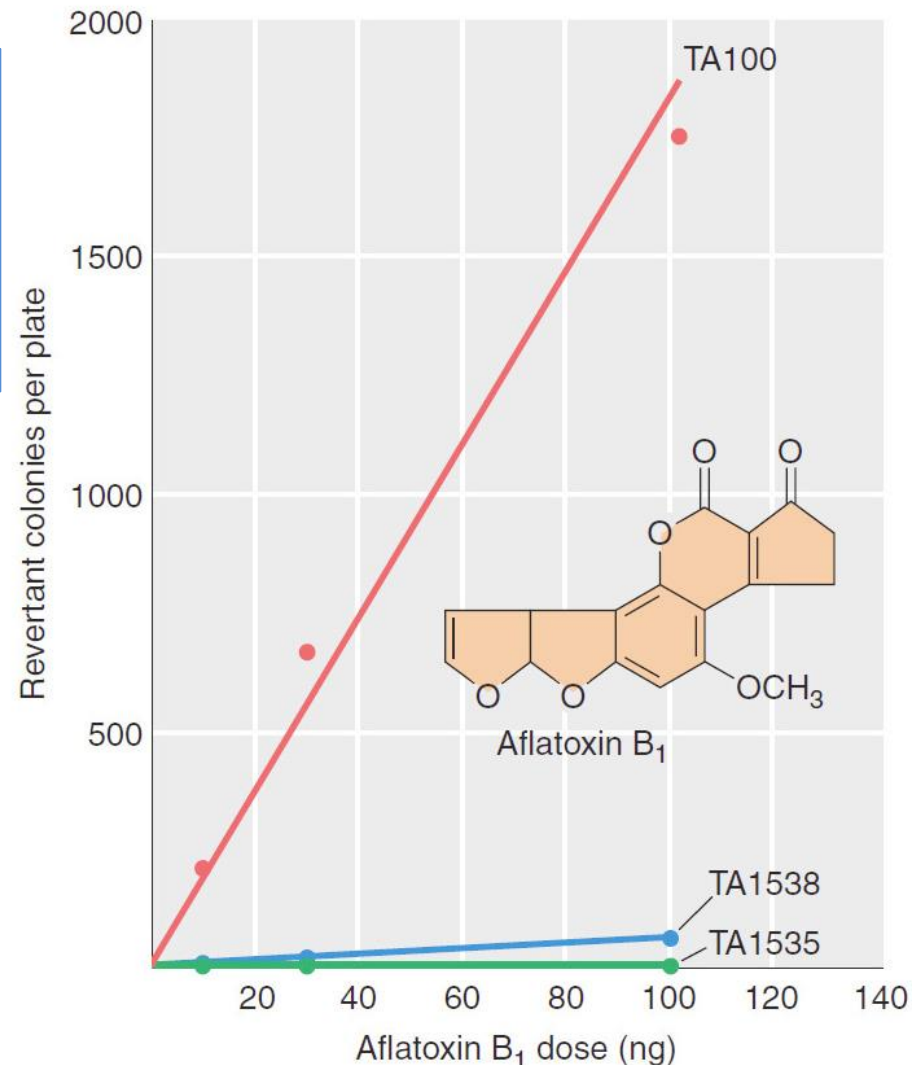
The Ames test for mutagenic compounds

use three different Salmonella strains
Strain 1 = His⁻ transition mutant
Strain 2,3 = His⁻ frameshift mutant
TA100 = His⁻ transition mutant
TA1535, TA 1538 = His⁻ frameshift mutant



Data interpretation:

What type of mutations
does Aflatoxin B₁ induce?



Data from J. McCann and B. N. Ames, in W. G. Flamm and M. A. Mehlman, eds. Advances in Modern Technology, vol. 5. Copyright by Hemisphere Publishing Corporation, Washington, D.C.